

On the effect of preload and pre-stretch on hemodynamic simulations: an integrative approach

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Abstract In this work, we address the simulation of three-dimensional arterial blood flow and its effect on the stress state of arterial walls. The novel contribution is the unprecedented combination of several modeling techniques to account for (1) the fact that known configurations for the arterial wall are in a preloaded state, (2) the compliance of the vessel segments, (3) proper boundary data over the non-physical interfaces resulting from the isolation of an arterial district from the rest of the arterial tree, (4) the presence of surrounding tissues in which the vessel is embedded and (5) residual stress state due to pre-stretch. Firstly, we formulate both the *forward* mechanical problem when the reference (zero-load) configuration is assumed to be known and, the *preload* problem arising when the known domain is a configuration at equilibrium with a certain load state (typically due to internal pressure and tethering forces). Then, two additional complexities are faced: the fluid–structure interaction problem that follows when the compliant vessels are coupled with the blood flow, and the introduction of non-physical

boundaries coming from the artificial isolation of the arterial district from the original vessel. This, in turn, poses the problem of coupling dimensionally heterogeneous models to incorporate the effect of upstream and downstream systemic impedances. Additionally, a viscoelastic support on the external surface of the vessel is also incorporated. Two examples are presented to quantify in a physiologically consistent scenario the differences in simulation results when either considering or not the preload state of arterial walls. These computational simulations shed light on the validity of simplifying hypotheses in most hemodynamic models.

Keywords Blood flow models · Preload configuration · Pre-stretch · Finite strains · 3D–1D modeling

1 Introduction

As far as cardiovascular diseases remain the leading cause of morbidity and mortality worldwide (Mendis et al. 2011; Heidenreich et al. 2011), both biomedical and scientific computing communities are being compelled to increase their efforts for further improving computational tools and simulation-based techniques to be used in the diagnosis and treatment of those pathologies. In the last decade, these communities have entered a virtuous cycle of reciprocal feedback, leading each time to a more realistic modeling of the complex biological structures and the associated physiological processes at all levels of biological organization (Southern et al. 2008). In particular, computational simulations are being used to assess causes, predict the development and optimize treatments and interventions of several pathologies in the cardiovascular system (CVS). This is due to the abilities and potentialities of such techniques in yielding relevant information, with improved level of detail and with better

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accuracy. Moreover, individualized patient-specific computational models are attractive for physicians and scientists, considering the potential benefits, the low cost implied in simulations and their associated impact on minimizing invasive procedures (Castro et al. 2011; Lewis and Kerckhoffs 2010).

However, computer models—more specifically computational biomechanics models and their associated mathematical formulations at the continuum level—are highly dependent on (1) the simplifying hypotheses about the interaction of the subsystems which are being considered, (2) the accuracy of the geometric representations of anatomical and functional structures and (3) the data that characterize the behavior of materials involved in the modeling process, for instance blood and the vascular soft tissues.

An adequate modeling of the entire arterial system (as an integrated functional unit) requires the identification of subsystems representing specific zones and/or particular functional aspects. In this analytic way, a mathematical model must be properly set for each subsystem in order to handle the underlying (biological, mathematical and numerical) complexity. Once mathematical models for these components are properly established, it is necessary to integrate them again into a set of coupled units to recreate the original dynamic interactions.

Focusing on the mechanical aspects of the blood circulation in the arterial system, accounting for truly realistic physiological conditions in blood flow simulations, implies in paying careful attention to the following aspects:

(a) Boundary conditions on non-physical interfaces When a portion of an artery is modeled via a full 3D physics, it is artificially isolated from the rest of the circulatory system. The simulation results within such arterial district are sensible to the coupling with the rest of the vascular tree. A suitable approach is to perform an embedding of the 3D model in a dimensionally reduced model (0D–1D for instance) of the entire arterial system to take into account global/local mutual interactions via consistent coupling conditions at the interfaces between these models. This issue was introduced in Formaggia et al. (1999) and Formaggia et al. (2001) and later exhaustively employed in Blanco et al. (2007, 2009, 2010), Urquiza et al. (2006), Vignon-Clementel et al. (2006), Migliavacca et al. (2006), Kim et al. (2009a).

(b) Constitutive modeling of the arterial wall The arterial wall can be considered as a multi-layered incompressible tissue that, from the mechanical point of view, is composed by an elastin matrix, collagen fibers and vascular smooth muscle cells. There exists consensus [see Holzapfel and Ogden (2010a) and references therein] that effects of elastin and collagen can be effectively modeled as a fiber rein-

forced hyperelastic material. Several constitutive equations have been proposed and validated against experimental data. Currently, the most extensively used constitutive model is the transversely isotropic model proposed in Holzapfel and Gasser (2000) and variations based on the constitutive equation introduced in this work [see Gasser et al. (2006) for example]. In addition, the passive response of the smooth muscle introduces viscoelastic effects that are characteristic of arterial tissues.

(c) Fluid–structure interaction Other relevant aspect to be considered is the adequate treatment of the fluid (blood) deformable domain due to the movement of the distensible arterial walls. It has been pointed out that in some cases the wall shear stress (WSS) may be quite overestimated if a rigid arterial wall assumption is considered in blood flow simulations (pure CFD modeling) (Hsu and Bazilevs 2011). This stresses the need for including the wall movement in flow simulations. Moreover, this issue goes beyond pure hydrodynamics, and FSI formulations must necessarily be considered if proper determination of stresses within the arterial tissue is required.

Fluid–structure interaction has been largely studied and a great deal of methods is available to tackle this problem, most of them based on the Arbitrary Lagrangian–Eulerian (ALE) formulation introduced in Hughes et al. (1981) and firstly applied in the context of hemodynamics by Taylor et al. (1998). Iterative and monolithic approaches have been proposed for the FSI system. In this regard, the literature is vast, and we limit to mention some relevant works in the field, e.g., Deparis et al. (2003), Deparis et al. (2006) for the former and Crosetto et al. (2011), Wu and Cai (2014), Joosten et al. (2009) for the latter.

(d) Preloaded configuration Actually, patient-specific computational models are based on in vivo geometrical structures obtained from medical image acquisition devices (like MRI and CT). Those images correspond to spatial configurations that are not load free. On the contrary, they are configurations which are equilibrated with the hemodynamic forces acting at the internal wall of the artery. In general, in the available literature this aspect is usually disregarded, and only in few publications (see Table 1) the initial image-based configuration is considered to be equilibrated by an inner baseline pressure. However, there is no clear exposition on how this pressure level is selected. A more realistic approach would be to estimate this inner pressure taking into account the data acquisition technology and the selected reconstruction algorithm; for example, in some cases, the given information is a mean over time of the positioning of the artery in the complete cardiac cycle (e.g., in 3D rotational angiographies (Orth et al. 2009; Rooij et al. 2008)), while in other studies, it is possible to synchronize the image acquisition with the

Table 1 Main contributions in modeling issues (see text for further explanation)

Article	Dimen. het. coupling	FSI	Finite strains	Histological model	Preloaded state	External tissue	Tethering stress
Grinberg and Kamiadakis (2008)	✓	–	–	–	–	–	–
Vignon-Clementel et al. (2006)	✓	–	–	–	–	–	–
Urquiza et al. (2006)	✓	(α)	–	–	–	–	–
Blanco et al. (2010)							
Blanco et al. (2013b)							
Torii et al. (2006)	–	✓	✓	–	–	–	–
Torii et al. (2008)							
Wu and Cai (2014)							
Lu et al. (2007)	–	–	✓	–	✓	–	–
Lu et al. (2008)							
Holzäpfel and Gasser (2000)	–	–	✓	✓	(β)	–	✓
Holzäpfel and Ogden (2010b)							
Crosetto et al. (2011)	–	✓	✓	–	–	✓	–
Reymond et al. (2013)							
Kim et al. (2009b)	✓	✓	✓	–	–	–	–
Kim et al. (2009a)							
Gee et al. (2009)	–	–	✓	✓	✓	–	–
Gee et al. (2010)							
Blanco et al. (2013a)	✓	✓	✓	–	–	✓	–
Malossi et al. (2013)							
Tezduyar et al. (2010)	–	✓	✓	✓	✓	–	–
Hsu and Bazilevs (2011)							
Vavourakis et al. (2011)							
Moireau et al. (2012)	–	✓	✓	–	✓	✓	–
Present work	✓	✓	✓	✓	✓	✓	✓

(α) Using simple 1D models for the arterial wall. (β) This work is based on ex-vivo measures, and the initial configurations are truly unloaded

patient's electrocardiogram, obtaining images corresponding to a constant time window within the cardiac pulse [e.g., on intravascular ultrasound studies (Mintz et al. 2001; Honda and Fitzgerald 2008)].

The most common approach to compute the stress state in this preloaded configuration is usually referred to as *inverse design analysis* (Govindjee and Mihalic 1996, 1998; Lu et al. 2008, 2007) and focuses on finding the zero-load configuration; a different approach is proposed in Gee et al. (2010, 2009) introducing *prestresses* and *pre-strains* in the image-based geometry.

A common criticism that can be extended to most articles dealing with this topic is that the biomechanical loads acting on the image-based configuration are not fully characterized. The presence of tethering stresses at the cross-sectional—artificial—boundaries is also usually disregarded, neglecting a factor that—as it is shown in Holzapfel and Ogden (2010b) and put in evidence in our numerical examples—plays a relevant role in determining the mechanical state of arterial walls.

It is also worthwhile to mention that, as stated by Gee et al. (2009), neglecting the fact that the image-based geometry corresponds to a loaded state and taking a simplified approach (considering them as load-free configurations being the most common case) leads to unrealistic evaluation of the quantities of interest such as strain, stress and other internal variables involved in the constitutive description. In this connection, it is useful to highlight the findings reported in Feng et al. (2007) related to the importance of aortic axial stretch for the correct operation of the baroreflex mechanotransducers that play an essential role in the proper control of the level of systemic arterial pressure.

(e) *Surrounding tissues* The arterial vessel is surrounded by tissues, which restrains the movement of the arterial wall, altering the deformation processes, and thence affecting the stress state. Most of the current FSI studies neglect this fact using an homogeneous Neumann boundary condition on such outer surface of the arterial wall, leading to unrealistic motions of the artery, as shown in Malossi et al. (2013). The main hindrance relies in the determination of the forces acting on the outer boundary. Furthermore, determining the outer boundary itself is hard as sometimes there is no clear distinction between adventitia and surrounding tissue. A recent article (Moireau et al. 2012) deals with this issue by modeling the external tissues as a viscoelastic medium. Reported results show improvements in the accuracy when comparing the predicted results (both considering and neglecting the surrounding tissue interaction) with experimental information extracted from medical images.

(f) *Residual stresses* Blood vessel tissues support a self-equilibrated stress state when they are at their load-free material configuration. Even with small magnitudes, stud-

ies suggest that these stresses have a significant influence on the stress state during the cardiac cycle (Holzapfel and Gasser 2000; Humphrey 2002). The presence of this residual stress is manifested when seizures are made to an arterial segment. For example, when the cut is performed axially, arteries show a retraction of its longitude and radial cuts result in an opening angle. To take into account this residual stress state, in Holzapfel and Gasser (2000) the authors consider as the reference (zero-stress) configuration a shortened and opened vessel geometry. The approximation of the residual stress state is given by the constitutive compatible stress field that arises when solving the equilibrium equations, “closing” the artery to its original in vivo topology and also stretching it to achieve its initial length. Moreover, great experimental and theoretical efforts have been directed to further characterize the residual stress state in Holzapfel et al. (2007) and Holzapfel and Ogden (2010b).

In this work, the effect of the axial stretch is taken into account, while the introduction of residual stresses due to the opening angle remains an open problem for non-cylindrical geometries. However, if the residual stress state was somehow provided, it can be straightforwardly integrated within the formulation presented in Sect. 2.

In Table 1, we present a summary of the modeling issues described above, and a contextualization of the major contributions of several research groups in the corresponding areas.

In view of the recent advances in each of the topics outlined above and the open field for research pointed out by Table 1, in the present work a modeling framework that integrates these aspects is proposed. This represents a step toward performing even more realistic hemodynamics simulations.

In summary, the strategies to be addressed here will include a full 3D model of a vascular district coupled with a lumped 1D–0D model of the rest of the CVS. The dimensionally heterogeneous coupling between models of reduced dimensions with full 3D zones is formulated in a consistent variational framework, following the developments described in Blanco et al. (2007). In particular, the 3D model accounts for the solution of the blood flow problem in a moving domain due to the wall compliance. The resulting FSI problem is treated with the ALE method for the fluid problem. The solid problem (arterial wall) is considered to be in the finite strain regime assuming a hyper-elastic material behavior, subjected to the hemodynamic loads exerted by the blood flow, a viscoelastic media surrounding the vessel and a tethering stress state acting at the artificial cross-sectional boundaries. The preload problem is also addressed, considering known a pre-strained configuration (for example, the diastolic geometry) and such problem consists in finding the load-free reference configuration. The tethering stress is also considered in the preload problem such that a certain level of pre-stretch is achieved. This fact will allow to take into

account a component of the residual stresses state, in a similar way as detailed in [Holzapfel and Ogden \(2010b\)](#). Moreover, the variational equilibrium problem for the solid problem is stated in a given deformed/loaded configuration considered to be extracted from medical images and the corresponding formulations are consistently linearized.

In this context, the contribution of the present work is twofold:

- (1) to present and discuss the conceptual background behind the variational formulations which are required in the hemodynamics field when a realistic mechanical analysis of arterial walls is sought, whether to solve the mechanical state associated with image-based arterial geometries or to solve the mechanical equilibrium for any other loading state during the entire cardiac cycle;
- (2) to introduce, for the first time, hemodynamic simulations integrating the items (a)–(e) discussed above, aiming to gain insight into the interplay among modeling ingredients and underlying hypotheses, and their impact on the assessment of the mechanical state of arterial walls.

We highlight that these two contributions are fundamental modeling issues in hemodynamic problems. Three numerical experiments are presented which are intended to show the viability of such integrative approach. Specifically, a sensitivity analysis is performed assessing the relevance of taking into account the complete load state acting over the arterial wall, evidencing the importance of considering both the internal pressure and tethering forces to appropriately characterize the mechanical response of the arterial wall. In order to facilitate the interpretation of the results and to avoid obscuring the analysis with intrinsic complexities present in patient-specific geometries, the first two numerical examples are performed using idealized arterial domains, namely a straight pipe and an ideal bifurcation. The last numerical example presents the application of the present modeling framework to a patient-specific geometry obtained from medical images, showing the feasibility and relevance of applying this techniques to realistic scenarios.

We remark that the realistic assessment of the stress state in arterial walls is of the utmost importance to feed models involving damage, remodeling and growth, being these phenomena the ultimate link to cardiovascular diseases.

The remainder of this document is organized as follows: Sect. 2 presents the mechanical formulation and linearization of the forward and preload solid problems, while in Sect. 3, the fluid problem is briefly described and the selected FSI strategy is detailed. A description of the dimensionally heterogeneous coupling is provided in Sect. 4. Section 5 presents some complementary modeling issues, such as the constitutive modeling and the numerical approximation

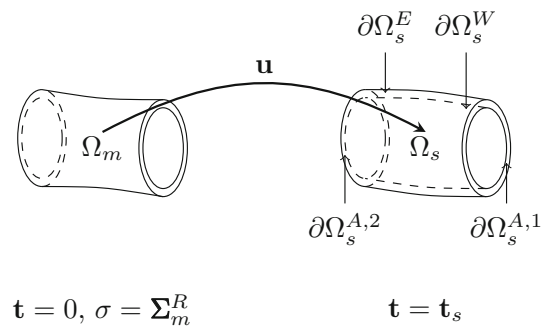


Fig. 1 Mechanical equilibrium setting

techniques. Section 6 presents the numerical experiments. Finally, results, limitations and some concluding remarks are outlined in Sect. 7.

2 Arterial wall mechanics

2.1 Kinematic settings

Consider the setting presented in Fig. 1, showing a body for which we identify in the Euclidean space its *spatial* configuration denoted by Ω_s with boundary $\partial\Omega_s = \partial\Omega_s^W \cup \partial\Omega_s^E \cup \partial\Omega_s^A$, where $\partial\Omega_s^W$ represents the interface between the vessel and the blood, $\partial\Omega_s^E$ the external surface, and $\partial\Omega_s^A = \bigcup_{i=1}^C \partial\Omega_s^{A,i}$ stands for the set of C cross-sectional—non-physical—boundaries. Points in this configuration are denoted by $\mathbf{x}_s$. A *material* configuration, used as a reference configuration, will be denoted by Ω_m , with points $\mathbf{x}_m$.

Remark 1 In the context of hemodynamics, Ω_s may refer to a diastolic, systolic or any other loaded state of the arterial wall. And it stands for the configuration at which mechanical equilibrium is achieved for a given load condition. We point out that the material configuration is always load free but not stress free if residual stresses are considered.

The displacement field from the reference to the spatial configuration is denoted by $\mathbf{u}$. Then, we can characterize the deformation mapping from Ω_m onto Ω_s and its inverse by the following expressions,

$$\mathbf{x}_s = \chi_m(\mathbf{x}_m) = \mathbf{x}_m + \mathbf{u}_m, \quad (1)$$

$$\mathbf{x}_m = \chi_s(\mathbf{x}_s) = \chi_m^{-1}(\mathbf{x}_s) = \mathbf{x}_s - \mathbf{u}_s, \quad (2)$$

where subscripts m and s denote the descriptions of the fields in the material and spatial configurations, respectively, that is, for the displacement vector field it is

$$\mathbf{u}_s(\mathbf{x}_s) = (\mathbf{u}_m(\mathbf{x}_m))_s = \mathbf{u}_m(\chi_m^{-1}(\mathbf{x}_s)). \quad (3)$$

This notation will extensively be used in the following. The gradients of these mappings are then given by

$$\mathbf{F}_m = \nabla_m \chi_m = \mathbf{I} + \nabla_m \mathbf{u}_m, \quad (4)$$

$$\mathbf{f}_s = \nabla_s \chi_s = \nabla_s \chi_m^{-1} = \mathbf{I} - \nabla_s \mathbf{u}_s, \quad (5)$$

where ∇_m and ∇_s denote the gradients with respect to material and spatial coordinates, respectively. Observe that $[\mathbf{F}_m^{-1}]_s = \mathbf{f}_s$ and $[\mathbf{f}_s^{-1}]_m = \mathbf{F}_m$.

Further, biological tissues behave as incompressible materials, this represents a kinematical condition that will be present in all subsequent developments, i.e.,

$$\det \mathbf{F}_m = 1. \quad (6)$$

2.2 Equilibrium equations for the arterial wall

In a general case, the load state of the arterial segment can be characterized as follows: a Neumann boundary condition given by the forces exerted by the blood flow over $\partial\Omega_s^W$, i.e., the traction $\mathbf{t}_s^W$, and a set of tethering tractions $\mathbf{t}_s^{A,i}$ acting over $\partial\Omega_s^{A,i}$, $i = 1, \dots, C$. Equivalently, we group the tethering tractions into $\mathbf{t}_s^A$, which is defined over the whole $\partial\Omega_s^A$. We will consider that the load exerted by external tissues $\mathbf{t}_s^E$ acting on the external surface $\partial\Omega_s^E$ is an active traction; that is, it depends on the solution of the problem (see Sect. 5.2). It is important to recall that the traction due to hemodynamic forces $\mathbf{t}_s^W$ constitutes a follower load and can be split into a normal and a tangential component by $\mathbf{t}_s^W = t_s^{W,n} \mathbf{n}_s + \mathbf{t}_s^{W,t}$ where $t_s^{W,n} = \mathbf{t}_s^W \cdot \mathbf{n}_s$ is the normal component of the traction and $\mathbf{t}_s^{W,t}$ is the tangent vector which can be characterized as $\mathbf{t}_s^{W,t} = \mathbf{P}_s \mathbf{t}_s^W$, with $\mathbf{P}_s = (\mathbf{I} - \mathbf{n}_s \otimes \mathbf{n}_s)$ being the tangential projection operator to the plane with unit normal vector $\mathbf{n}_s$. Note that we can trivially write $\mathbf{t}_s^{W,t} = \mathbf{P}_s \mathbf{t}_s^{W,t}$.

The momentum conservation equation is presented considering the incompressibility as an additional constraint. Then, an independent variable (the pressure in the solid domain) emerges as a Lagrange multiplier to accommodate this kinematic constraint.

Next, we provide the statement of two variational formulations of mechanical equilibrium whose difference is rather subtle: the known domain. In Problem 1 below, the given domain is the domain at which the body is at equilibrium (the spatial domain), and the unknown domain is the reference domain used to define the constitutive equations (the material domain). In Problem 2, the given domain is the material domain, while the unknown data are the spatial domain.

Problem 1 (Preload problem) Given the equilibrium configuration Ω_s and the load acting on the spatial configuration $(\mathbf{t}_s^W, \mathbf{t}_s^E, \mathbf{t}_s^A)$, the variational equations that govern this problem are those corresponding to the mechanical equilibrium expressed in the *spatial* domain. Hence, the problem reads: given the loading system $(t_s^{W,n}, \mathbf{t}_s^{W,t}, \mathbf{t}_s^E, \mathbf{t}_s^A)$, find $(\mathbf{u}_s, \lambda_s) \in \mathcal{U}_s \times \mathcal{L}_s$ such that σ_s satisfies

$$\begin{aligned} & \int_{\Omega_s} [-\lambda_s \operatorname{div} \hat{\mathbf{u}}_s + \sigma_s \cdot \boldsymbol{\varepsilon}_s(\hat{\mathbf{u}}_s)] \, d\Omega_s \\ &= \int_{\partial\Omega_s^W} \mathbf{P}_s \mathbf{t}_s^{W,t} \cdot \hat{\mathbf{u}}_s \, d\partial\Omega_s^W + \int_{\partial\Omega_s^W} t_s^{W,n} \mathbf{n}_s \cdot \hat{\mathbf{u}}_s \, d\partial\Omega_s^W \\ & \quad + \int_{\partial\Omega_s^E} \mathbf{t}_s^E \cdot \hat{\mathbf{u}}_s \, d\partial\Omega_s^E + \sum_{i=1}^C \int_{\partial\Omega_s^{A,i}} \mathbf{t}_s^{A,i} \cdot \hat{\mathbf{u}}_s \, d\partial\Omega_s^{A,i} \\ & \quad \forall \hat{\mathbf{u}}_s \in \mathcal{V}_s, \end{aligned} \quad (7)$$

$$\int_{\Omega_s} [1 - \det \mathbf{F}_s^{-1}] \hat{\lambda}_s \, d\Omega_s = 0 \quad \forall \hat{\lambda}_s \in \mathcal{L}_s, \quad (8)$$

where $\boldsymbol{\varepsilon}_s(\hat{\mathbf{u}}) = \frac{1}{2}(\nabla_s \hat{\mathbf{u}} + \nabla_s^T \hat{\mathbf{u}})$ is the strain rate tensor in the *spatial* configuration, $\mathcal{L}_s = L^2(\Omega_s)$ and $\mathcal{U}_s = \{\mathbf{u}_s \in \mathbf{H}^1(\Omega_s), \mathbf{u}_s \text{ satisfies essential b.c.}\}$ are, respectively, the linear space for pressures and the linear manifold associated with kinematically admissible displacements, and $\mathcal{V}_s = \{\hat{\mathbf{u}}_s \in \mathbf{H}^1(\Omega_s), \hat{\mathbf{u}}_s \text{ satisfies null essential b.c.}\}$ is the space of kinematically admissible variations. Also, σ_s can be related to the second Piola–Kirchhoff stress tensor $\mathbf{S}_m$ through

$$\sigma_s = \frac{1}{\det \mathbf{F}_s} \mathbf{F}_s \left(\boldsymbol{\Sigma}_m^R + \mathbf{S}_m(\mathbf{E}_m) \right) \mathbf{F}_s^T. \quad (9)$$

where $\boldsymbol{\Sigma}_m^R$ is a known residual stress field defined in the material configuration and the Piola–Kirchhoff stress tensor is a function of the Green–Lagrange deformation tensor $\mathbf{E}_m = \frac{1}{2}(\mathbf{F}_m^T \mathbf{F}_m - \mathbf{I})$ via a constitutive equation (see Sect. 5.1).

Remark 2 The modeling of residual stresses is currently a challenging open problem, and current efforts are based on the analysis of ex-vivo data (Holzapfel and Gasser 2000; Alastrué et al. 2007; Holzapfel and Ogden 2010b). However, there is not yet available a methodology to systematically introduce residual stresses on arbitrary geometries. Considering that in the future the residual stress could be a given data; in the present work, this field is included in the mathematical developments. Residual stresses are usually separated into two different kinds, namely: those related to the opening-angle phenomenon, and those related to axial stretch. In this work, we consider tethering forces which account for the latter kind.

Problem 2 (Forward problem) When the material (load free but not stress free due to residual stresses) configuration Ω_m is known, the variational Eqs. (7)–(8) can be evaluated in terms of this reference domain, yielding what is known as the *forward problem*. In fact, the variational equations that govern the equilibrium problem expressed now in the *material* domain Ω_m are written as follows: given the material description of the load system, $(t_m^{W,n}, \mathbf{t}_m^{W,t}, \mathbf{t}_m^E, \mathbf{t}_m^A)$, find $(\mathbf{u}_m, \lambda_m) \in \mathcal{U}_m \times \mathcal{L}_m$ such that

$$\begin{aligned}
& - \int_{\Omega_m} \lambda_m \left(\mathbf{F}_m^{-T} \cdot \nabla_m \hat{\mathbf{u}}_m \right) \det \mathbf{F}_m \, d\Omega_m \\
& + \int_{\Omega_m} (\boldsymbol{\Sigma}_m^R + \mathbf{S}_m(\mathbf{E}_m)) \cdot \dot{\mathbf{E}}(\hat{\mathbf{u}}_m) \, d\Omega_m \\
& = \int_{\partial\Omega_m^W} \left(\mathbf{t}_m^{W,n} \mathbf{F}_m^{-T} \mathbf{n}_0^W \cdot \hat{\mathbf{u}}_m \right) \det \mathbf{F}_m \, d\partial\Omega_m^W \\
& + \int_{\partial\Omega_m^W} \mathbf{P}_m \mathbf{t}_m^{W,t} \cdot \hat{\mathbf{u}}_m | \mathbf{F}_m^{-T} \mathbf{n}_0^W | \det \mathbf{F}_m \, d\partial\Omega_m^W \\
& + \int_{\partial\Omega_m^E} \left(\mathbf{t}_m^E \cdot \hat{\mathbf{u}}_m \right) | \mathbf{F}_m^{-T} \mathbf{n}_0^E | \det \mathbf{F}_m \, d\partial\Omega_m^E \\
& + \sum_{i=1}^C \int_{\partial\Omega_m^{A,i}} \left(\mathbf{t}_m^{A,i} \cdot \hat{\mathbf{u}}_m \right) | \mathbf{F}_m^{-T} \mathbf{n}_0^{A,i} | \det \mathbf{F}_m \, d\partial\Omega_m^{A,i} \\
& \quad \forall \hat{\mathbf{u}}_m \in \mathcal{V}_m, \quad (10) \\
& \int_{\Omega_m} (\det \mathbf{F}_m - 1) \hat{\lambda}_m \, d\Omega_m = 0 \quad \forall \hat{\lambda}_m \in \mathcal{L}_m, \quad (11)
\end{aligned}$$

where $\dot{\mathbf{E}}(\hat{\mathbf{u}}_m) = \frac{1}{2} [\mathbf{F}_m^T (\nabla_m \hat{\mathbf{u}}_m) + (\nabla_m \hat{\mathbf{u}}_m)^T \mathbf{F}_m]$, $\mathbf{n}_0$ is the unit normal vector in the material configuration, $\mathbf{P}_m$ is the projection operator in material description, which is given by

$$\mathbf{P}_m = \mathbf{I} - \frac{\mathbf{F}_m^{-T} \mathbf{n}_0^W}{| \mathbf{F}_m^{-T} \mathbf{n}_0^W |} \otimes \frac{\mathbf{F}_m^{-T} \mathbf{n}_0^W}{| \mathbf{F}_m^{-T} \mathbf{n}_0^W |}, \quad (12)$$

and $\mathcal{U}_m$, $\mathcal{V}_m$ and $\mathcal{L}_m$ are the counterparts of $\mathcal{U}_s$, $\mathcal{V}_s$ and $\mathcal{L}_s$, respectively, with functions defined in Ω_m .

We emphasize that the inertial term could be easily incorporated in the solid problem. However, for the arterial districts investigated in Sect. 6, the stresses associated with inertial forces could be disregarded without loss of accuracy. This is easily justified by a straightforward non-dimensionalization of the equations (for an essential cylindrical geometry), which shows that the non-dimensional number α_{in} , representing the ratio between inertial and circumferential stresses arising within the cardiac cycle is of the order of 10^{-6} . In fact, α_{in} can be approximated by

$$\alpha_{in} = \frac{\rho \Delta r_i h}{T^2 p_i}, \quad (13)$$

where r_i denotes the inner radius, h the thickness, p_i the internal pressure, Δr_i the inner radius variation, ρ the density of the wall and T the time it takes to achieve the peak pressure from diastole. Given the following values within the physiological range, $r_i = 0.4[\text{cm}]$, $\Delta r_i = 0.2r_i$, $h = 0.2r_i$, $p_i = 15 \cdot 10^4 [\frac{\text{dyn}}{\text{cm}^2}]$, $\rho = 1 [\frac{\text{gr}}{\text{cm}^3}]$ and $T = 0.1[\text{s}]$, through Eq. (13) $\alpha_{in} = 4.267 \cdot 10^{-6}$ is obtained.

This also holds for the *preload problem*, but in order to account for the inertial term, the velocity of the body at the equilibrium configuration should be also given. Nevertheless, the inertial term will be negligible due to the image-based geometries being corresponding to diastolic relaxed states with low velocities and accelerations.

Remark 3 The *forward problem* stated above is the classical equilibrium problem in solid mechanics when the material configuration is known, while the preload problem is a non-traditional formulation of the very same variational problem when the equilibrated (deformed) configuration is known. Therefore, both mechanical problems are governed by the same equilibrium equation, and the difference simply lies in the data for the problem and therefore in the corresponding linearized expressions.

Remark 4 The *preload problem* stated above is a mandatory step toward characterizing the mechanical state, i.e., the stress state, of the arterial wall at the baseline geometry (e.g., the end-diastolic geometry) with a given baseline hemodynamics loads (e.g., the end-diastolic pressure). The material configuration is required because it is used to define constitutive equations. As mentioned before, such baseline geometry is usually obtained from medical images, while the baseline hemodynamics loads can be easily estimated either from measurements or from the domain knowledge. Just after solving the preload problem, the baseline mechanical state (the stress state due to preload) is adequately determined and the *forward problem* involving fluid–structure interaction and dimensionally heterogeneous coupling can be solved to determine the equilibrium configuration for other hemodynamics loads occurring during the cardiac cycle. This is the reason why we need the use of the two forms associated with the evaluation of the mechanical equilibrium. Note that this problem is solved only once, for a given set of loads, and then, the *forward problem* is solved with a time-evolving set of loads given by the action of the blood flow during the cardiac cycle driven by the pulsation of the heart imposed through the coupling with the dimensionally reduced model of the CVS.

2.3 Linearization procedure of the equilibrium problem

For the two equilibrium (generally nonlinear) problems setup in Sect. 2.2, i.e., (1) the mechanical equilibrium when the material configuration Ω_m is known and (2) the mechanical equilibrium when the spatial configuration Ω_s is known, the corresponding linearized forms via the Newton method are derived in “Appendix 2”. Furthermore and for the sake of simplicity, in these derivations, we restrict our presentation to the evaluation of the contribution of the follower loads ($\mathbf{t}_s^{W,n}$ and $\mathbf{t}_s^{W,t}$) and a general static load $\mathbf{t}_s^N$ acting on a boundary $\partial\Omega_s^N$. Also, the known residual stress field $\boldsymbol{\Sigma}_m^R$ is neglected. The

generalization of this procedure for the complete problem is straightforward.

2.3.1 Newton–Raphson method for Problem 1 (Ω_s known)

We apply the Newton–Raphson linearization procedure to Eqs. (7)–(8), and for each iteration, a new material configuration Ω_m^k is obtained, with points $\mathbf{x}_m^k = \mathbf{x}_s - \mathbf{u}_s^k$. For the sake of clarity, we omit index k on the displacement field $\mathbf{u}_s$ (and also on quantities that are updated on each step, such as $\mathbf{F}_s$). Thus, the linear problem reads: given $(\mathbf{u}_s, \lambda_s)$ (displacement and pressure fields at previous Newton–Raphson iteration) find $(\delta\mathbf{u}_s, \delta\lambda_s) \in \mathcal{V}_s \times \mathcal{L}_s$ such that

$$\begin{cases} d_s(\delta\mathbf{u}_s, \hat{\mathbf{u}}_s) + e_s(\delta\lambda_s, \hat{\mathbf{u}}_s) = g_s(\hat{\mathbf{u}}_s) & \forall \hat{\mathbf{u}}_s \in \mathcal{V}_s \\ f_s(\delta\mathbf{u}_s, \hat{\lambda}_s) = h_s(\hat{\lambda}_s) & \forall \hat{\lambda}_s \in \mathcal{L}_s \end{cases} \quad (14)$$

with

$$\begin{aligned} d_s(\delta\mathbf{u}_s, \hat{\mathbf{u}}_s) = & - \int_{\Omega_s} (\mathbf{F}_s^T \cdot \nabla_s \delta\mathbf{u}_s) \boldsymbol{\sigma}_s \cdot \boldsymbol{\varepsilon}_s(\hat{\mathbf{u}}_s) d\Omega_s \\ & + \int_{\Omega_s} 2(\mathbf{F}_s(\nabla_s \delta\mathbf{u}_s) \boldsymbol{\sigma}_s) \cdot \boldsymbol{\varepsilon}_s(\hat{\mathbf{u}}_s) d\Omega_s \\ & + \int_{\Omega_s} \mathbf{D}_s(\mathbf{F}_s \nabla_s \delta\mathbf{u}_s)^S \cdot \boldsymbol{\varepsilon}_s(\hat{\mathbf{u}}_s) d\Omega_s, \end{aligned} \quad (15)$$

$$e_s(\delta\lambda_s, \hat{\mathbf{u}}_s) = - \int_{\Omega_s} \delta\lambda_s \operatorname{div} \hat{\mathbf{u}}_s d\Omega_s, \quad (16)$$

$$\begin{aligned} g_s(\hat{\mathbf{u}}_s) = & - \int_{\Omega_s} [-\lambda_s \operatorname{div} \hat{\mathbf{u}}_s + \boldsymbol{\sigma}_s \cdot \boldsymbol{\varepsilon}_s(\hat{\mathbf{u}}_s)] d\Omega_s \\ & + \int_{\partial\Omega_s^W} \mathbf{P}_s \mathbf{t}_s^W \cdot \hat{\mathbf{u}}_s d\partial\Omega_s^W + \int_{\partial\Omega_s^W} t_s^{W,n} \mathbf{n}_s \cdot \hat{\mathbf{u}}_s d\partial\Omega_s^W \\ & + \int_{\partial\Omega_s^N} \mathbf{t}_s^N \cdot \hat{\mathbf{u}}_s d\partial\Omega_s^N \end{aligned} \quad (17)$$

$$f_s(\delta\mathbf{u}_s, \hat{\lambda}_s) = - \int_{\Omega_s} (\mathbf{F}_s^T \cdot \nabla_s \delta\mathbf{u}_s) \hat{\lambda}_s d\Omega_s \quad (18)$$

$$h_s(\hat{\lambda}_s) = - \int_{\Omega_s} [1 - \det \mathbf{F}_s^{-1}] \hat{\lambda}_s d\Omega_s \quad (19)$$

2.3.2 Newton–Raphson method for Problem 2 (Ω_m known)

We apply the Newton–Raphson linearization procedure to Eqs. (10)–(11) and write the expressions in the configuration found in the last iteration. Observe that the configuration is updated to the current deformed domain where, when convergence is achieved, exact equilibrium is satisfied. As before, we omit index k indicating the quantities at the last iteration.

Thus, the linear problem reads: given $(\mathbf{u}_s, \lambda_s)$ (displacement and pressure fields at previous Newton–Raphson iteration) find $(\delta\mathbf{u}_s, \delta\lambda_s) \in \mathcal{V}_s \times \mathcal{L}_s$ such that

$$\begin{cases} a_s(\delta\mathbf{u}_s, \hat{\mathbf{u}}_s) + b_s(\delta\lambda_s, \hat{\mathbf{u}}_s) = l_s(\hat{\mathbf{u}}_s) & \forall \hat{\mathbf{u}}_s \in \mathcal{V}_s \\ c_s(\delta\mathbf{u}_s, \hat{\lambda}_s) = m_s(\hat{\lambda}_s) & \forall \hat{\lambda}_s \in \mathcal{L}_s \end{cases} \quad (20)$$

where the bilinear and linear forms are given by

$$\begin{aligned} a_s(\delta\mathbf{u}_s, \hat{\mathbf{u}}_s) = & \int_{\Omega_s} [\mathbf{D}_s + \lambda_s (2\mathbb{I} - (\mathbf{I} \otimes \mathbf{I}))] \boldsymbol{\varepsilon}(\delta\mathbf{u}_s) \cdot \boldsymbol{\varepsilon}(\hat{\mathbf{u}}_s) d\Omega_s \\ & + \int_{\Omega_s} (\nabla_s \delta\mathbf{u}_s) \boldsymbol{\sigma}_s \cdot (\nabla_s \hat{\mathbf{u}}_s) d\Omega_s \\ & + \int_{\partial\Omega_s^W} [t_s^{W,n} ((\nabla_s \delta\mathbf{u}_s)^T - \mathbf{I}) \mathbf{n}_s^W \cdot \hat{\mathbf{u}}_s] \operatorname{div}_s \delta\mathbf{u}_s d\partial\Omega_s^W \\ & - \int_{\partial\Omega_s^W} (\mathbf{H}(\delta\mathbf{u}_s) \mathbf{n}_s \otimes \mathbf{n}_s)^S \mathbf{t}_s^{W,t} \cdot \hat{\mathbf{u}}_s d\partial\Omega_s^W \\ & - \int_{\partial\Omega_m^W} \mathbf{P}_s \mathbf{t}_s^{W,t} \cdot \hat{\mathbf{u}}_s (\operatorname{div}_s \delta\mathbf{u}_s) d\partial\Omega_s^W \\ & - \int_{\partial\Omega_m^W} \mathbf{P}_s \mathbf{t}_s^{W,t} \cdot \hat{\mathbf{u}}_s \left(-(\nabla_s \delta\mathbf{u}_s)^T \mathbf{n}_s \cdot \mathbf{n}_s \right) d\partial\Omega_s^W \\ & - \int_{\partial\Omega_s^N} (\mathbf{t}_s^N \cdot \hat{\mathbf{u}}_s) \left[(\operatorname{div}_s \delta\mathbf{u}_s) - (\nabla_s \delta\mathbf{u}_s)^T \mathbf{n}_s \cdot \mathbf{n}_s \right] d\partial\Omega_s^N, \end{aligned} \quad (21)$$

$$b_s(\delta\lambda_s, \hat{\mathbf{u}}_s) = - \int_{\Omega_s} \delta\lambda_s \operatorname{div} \hat{\mathbf{u}}_s d\Omega_s, \quad (22)$$

$$\begin{aligned} l_s(\hat{\mathbf{u}}_s) = & - \int_{\Omega_s} [-\lambda_s \operatorname{div} \hat{\mathbf{u}}_s + \boldsymbol{\sigma}_s \cdot \boldsymbol{\varepsilon}_s(\hat{\mathbf{u}}_s)] d\Omega_s \\ & + \int_{\partial\Omega_s^W} \mathbf{P}_s \mathbf{t}_s^W \cdot \hat{\mathbf{u}}_s d\partial\Omega_s^W + \int_{\partial\Omega_s^W} t_s^{W,n} \mathbf{n}_s \cdot \hat{\mathbf{u}}_s d\partial\Omega_s^W \\ & + \int_{\partial\Omega_s^N} \mathbf{t}_s^N \cdot \hat{\mathbf{u}}_s d\partial\Omega_s^N, \end{aligned} \quad (23)$$

$$c_s(\delta\mathbf{u}_s, \hat{\lambda}_s) = \int_{\Omega_s} \operatorname{div} \delta\hat{\mathbf{u}}_s \hat{\lambda}_s d\Omega_s, \quad (24)$$

and

$$m_s(\hat{\lambda}_s) = - \int_{\Omega_s} (1 - \det \mathbf{F}_s^{-1}) \hat{\lambda}_s d\Omega_s. \quad (25)$$

where $\mathbf{H}(\delta\mathbf{u}_s)$ and $\mathbf{D}_s \boldsymbol{\varepsilon}_s(\delta\mathbf{u}_s)$ stand for

$$\mathbf{H}(\delta\mathbf{u}_s) = \left[-(\nabla_s \delta\mathbf{u}_s)^T + \left((\nabla_s \delta\mathbf{u}_s)^T \mathbf{n}_s \cdot \mathbf{n}_s \right) \mathbf{I} \right] \quad (26)$$

and

$$\mathbf{D}_s \boldsymbol{\varepsilon}_s(\delta \mathbf{u}_s) = \frac{1}{\det \mathbf{F}_s} \mathbf{F}_s \left[\left(\frac{\partial \mathbf{S}_m}{\partial \mathbf{E}_m} \right)_s \mathbf{F}_s^T \boldsymbol{\varepsilon}(\delta \mathbf{u}_s) \mathbf{F}_s \right] \mathbf{F}_s^T, \quad (27)$$

respectively. The components in a Cartesian system for the fourth-order tensor $\mathbf{D}_s$ are detailed in “Appendix 2” by Eq. (65). Recall that all integrals in (21)–(25) are in the known configuration at previous Newton–Raphson iteration.

3 Fluid–structure interaction

3.1 Setting for ALE formulation

The ALE formulation is particularly useful in flow problems involving large distortions in the presence of mobile and deforming boundaries (Donea and Huerta 2003), e.g., in FSI problems with finite displacements. Within this framework, a referential deformable domain is introduced and the fluid motion is related to points in that volume.

Consider the kinematic setting shown in Fig. 2. Let γ_t be the domain representing the lumen of the vessel at a given time t , and γ_0 a reference configuration of this domain at $t = t_0$.

Considering $\tau \in \{0, t\}$, the boundary is defined by $\partial \gamma_\tau = \partial \gamma_\tau^{A,i} \cup \partial \gamma_\tau^W$, $i = 1, \dots, C$, where $\partial \gamma_\tau^{A,i}$ is the i th inflow/outflow—artificial—boundary, $i = 1, \dots, C$, being C the number of inflow/outflow artificial boundary resulting from isolating the geometry from the rest of the system. The interface with the arterial wall is $\partial \gamma_\tau^W$ (fluid–solid interface). Points in the mentioned configurations are denoted by $\mathbf{x}_t$ and $\mathbf{x}_0$, accordingly.

Within this section, $\mathbf{v}$ and p are employed to denote blood velocity and pressure, respectively, and $\mathbf{d}$ denotes the displacement of the domain γ_t with respect to the reference configuration γ_0 . All these fields are defined in the spatial domain γ_t .

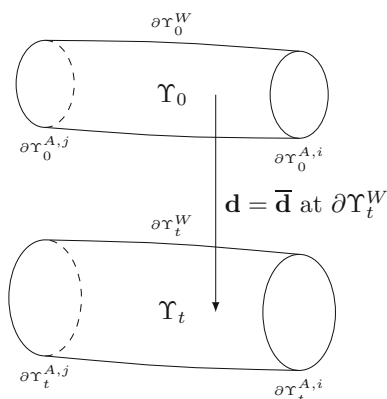


Fig. 2 ALE kinematic setting

The displacement (and velocity) of the fluid–solid interface $\partial \gamma_t^W$ equals the displacement of the arterial wall. This is the kinematic coupling condition for the fluid–structure interaction problem, that is

$$\mathbf{d}|_{\partial \gamma_t^W} = \bar{\mathbf{d}}_t, \quad (28)$$

where $\bar{\mathbf{d}}_t$ is given by the displacement of the arterial wall at the fluid–solid interface. If the reference configuration γ_0 is chosen to be the lumen enclosed within an image-based configuration (say Ω_I , with displacements $\mathbf{u}^I$), then it follows that

$$\bar{\mathbf{d}}_t = \mathbf{u}_s|_{\partial \Omega_s^W} - \mathbf{u}_s^I|_{\partial \Omega_s^W}, \quad (29)$$

where $\mathbf{u}_s|_{\partial \Omega_s^W}$ denotes the solid displacements at the fluid–structure interface for a given time t .

3.2 Equilibrium equations for the fluid problem

Here the momentum and mass conservation equations for the flow of an incompressible fluid in a compliant domain expressed in the spatial configuration are presented. Since all the quantities are expressed in terms of $\mathbf{x}_t$ coordinates, subscripts are omitted to simplify the notation. No-slip boundary conditions are imposed on $\partial \gamma_t^W$ for the velocity field. Neumann boundary conditions on the artificial boundaries $\partial \gamma_t^{A,i}$, $i = 1, \dots, C$, are denoted by $\mathbf{t}_t^i$ and correspond to the action of the rest of the cardiovascular system (CVS) over such boundaries. This coupling is addressed in Sect. 4. The traction $\mathbf{t}_t^i$ is actually assumed to be a vector in the direction of the normal vector $\mathbf{n}^i$, so we write $\mathbf{t}_t^i = t_t^i \mathbf{n}^i$, where t_t^i is the magnitude of the traction vector and $\mathbf{n}^i$ is the outward unit normal of $\partial \gamma_t^{A,i}$, $i = 1, \dots, C$. Loads per unit volume are neglected.

For this setting, the variational problem reads: for every $t \in [0, T]$ find $(\mathbf{v}, p, \mathbf{d}) \in \mathcal{V}_t \times \mathcal{P}_t \times \mathcal{D}_t$ such that

$$\begin{aligned} & \int_{\gamma_t} \rho \frac{\partial \mathbf{v}}{\partial t} \cdot \hat{\mathbf{v}} \, d\gamma_t + \int_{\gamma_t} \rho \nabla \mathbf{v} (\mathbf{v} - \mathbf{w}) \cdot \hat{\mathbf{v}} \, d\gamma_t \\ & - \int_{\gamma_t} p \operatorname{div} \hat{\mathbf{v}} \, d\gamma_t + \int_{\gamma_t} \bar{\boldsymbol{\sigma}}(\mathbf{v}) \cdot \boldsymbol{\varepsilon}(\hat{\mathbf{v}}) \, d\gamma_t \\ & = \sum_{i=1}^C \int_{\partial \gamma_t^{A,i}} t_t^i \mathbf{n}^i \cdot \hat{\mathbf{v}} \, d\partial \gamma_t^{A,i} \quad \forall \hat{\mathbf{v}} \in \mathcal{V}_t^*, \end{aligned} \quad (30)$$

$$\int_{\gamma_t} \hat{p} \operatorname{div} \mathbf{v} \, d\gamma_t = 0 \quad \forall \hat{p} \in \mathcal{P}_t, \quad (31)$$

$$\int_{\gamma_t} \nabla \mathbf{d} \cdot \nabla \hat{\mathbf{d}} \, d\gamma_t = 0 \quad \forall \hat{\mathbf{d}} \in \mathcal{D}_t^*, \quad (32)$$

where ρ is the fluid density, σ is such that $\sigma = -p\mathbf{I} + \bar{\sigma}$, which for a Newtonian incompressible fluid we consider $\bar{\sigma} = 2\mu\epsilon(\mathbf{v})$, μ is the dynamic viscosity and the velocity of the frame of reference is $\mathbf{w} = \left(\frac{\partial \mathbf{d}_0}{\partial t}\right)_t = \mathbf{F}_t^A \frac{\partial \mathbf{d}_t}{\partial t}$, where $\mathbf{F}_0^A = \frac{\partial \mathbf{x}_t}{\partial \mathbf{x}_0}$. Also, it is given by

$$\begin{aligned}\mathcal{V}_t &= \left\{ \mathbf{v} \in \mathbf{H}^1(\mathcal{V}_t), \mathbf{v}|_{\partial\mathcal{V}_t^W} = \left(\frac{\partial \bar{\mathbf{d}}_0}{\partial t}\right)_t \right\} \\ \mathcal{P}_t &= L^2(\mathcal{V}_t) \\ \mathcal{D}_t &= \left\{ \mathbf{d} \in \mathbf{H}^1(\mathcal{V}_t), \mathbf{d}|_{\partial\mathcal{V}_t^W} = \bar{\mathbf{d}}_t \right\}\end{aligned}\quad (33)$$

and $\mathcal{V}_t^*$ and $\mathcal{D}_t^*$ are the corresponding spaces of kinematically admissible variations obtained from taking differences of elements of the corresponding linear manifolds. A linearization for these equations using a fixed-point method is detailed in “Appendix 2”.

3.3 Interface equations

Consider at the given time t , the arterial wall is at equilibrium in a configuration Ω_s . At the interface $\partial\Omega_s^W = \partial\mathcal{V}_t^W$, the following coupling equations hold

$$\begin{aligned}\mathbf{u}_s &= \mathbf{d} && \text{at } \partial\Omega_s^W, \\ (\sigma_s - \sigma)\mathbf{n}_s &= 0 && \text{at } \partial\Omega_s^W,\end{aligned}\quad (34)$$

with $\mathbf{n}_s$ representing the normal vector at the interface.

3.4 Coupling algorithm

The FSI problem is solved within the context of a dimensionally heterogeneous coupling algorithm (3D–1D coupling, see Sect. 4). Thus, it is actually a black-box component from the point of view of the 3D–1D coupling algorithm. In fact, as time evolves, computations are carried out with the following strategy:

Solve preload problem (Section 2)
 Loop in time
 Loop for 3D–1D coupling (Broyden iterations, Section 4)
 Data (D) sent to components
 Loop for FSI coupling (fixed-point method, see next)
 Outcome (O) retrieved from components

The data for the *FSI black-box component* are

- (D.i) at each time step: quantities from previous time step (fluid quantities);
- (D.ii) at each 3D–1D coupling iteration: the normal component of the traction vectors t_t^i at $\partial\mathcal{V}_t^{A,i}$, $i =$

- $1, \dots, C$, which are provided by the 3D–1D algorithm, and that are incorporated as Neumann boundary conditions on the fluid problem [see Eq. (30)];
- (D.iii) at each FSI coupling iteration: quantities at previous iteration (fluid and solid quantities).

Solving the FSI problem implies solving the fully coupled system of equations given by (10)–(11) (arterial wall mechanics) and (30)–(32) (blood flow dynamics).

As outcome, the solution of the FSI problem provides the 3D–1D coupling algorithm with the flow rates at each non-physical boundary given by

$$Q_i = \int_{\partial\mathcal{V}_t^{A,i}} \mathbf{v} \cdot \mathbf{n}^i \, d\partial\mathcal{V}_t^{A,i} \quad i = 1, \dots, C. \quad (35)$$

Figure 3 presents the FSI algorithm based on fixed-point iterations. The FSI coupling loop corresponds to the circuit of blocks labeled with letters from (a) to (g). Blocks in yellow correspond to the input/output connections with the 3D–1D coupling algorithm. The key blocks of the FSI scheme are the components (b) and (f), standing for one iteration of the fixed-point scheme of the fluid problem [see Eqs. (98)–(100) in “Appendix 2”] and one step of the Newton–Raphson linearization of the solid—forward—problem [system of equations given by (20)]. Summarizing, the fluid–structure algorithm reads:

Loop for FSI iterations
 Define boundary conditions for fluid problem (a)
 Solve 1 iteration for linearized fluid problem (b)
 Subrelaxation step κ_a (c)
 Define boundary conditions for solid problem (d)–(e)
 Solve 1 iteration for linearized solid problem (f)
 Subrelaxation step κ_b (g)
 Evaluate convergence (h)

In order to avoid the *overshooting* phenomena, a further relaxation is introduced in the definition of the Neumann boundary condition for the inner surface of the solid (block (e)) through $\mathbf{t}_s^W = \iota \cdot \mathbf{t}_s^{W,k} + (1 - \iota) \cdot \mathbf{t}_s^{W,k-1}$, where $\iota = k/n_s$ if $k < n_s$. This way, the load is applied gradually in the first n_s (problem dependent parameter) steps. For the examples detailed in Sect. 6, $n_s = 6$ resulted in a trade-off between robustness and computational costs.

4 Dimensionally heterogeneous models

4.1 Hemodynamic components

The 3D FSI model, in which a detailed description of the hemodynamics is sought, is embedded into a larger system,

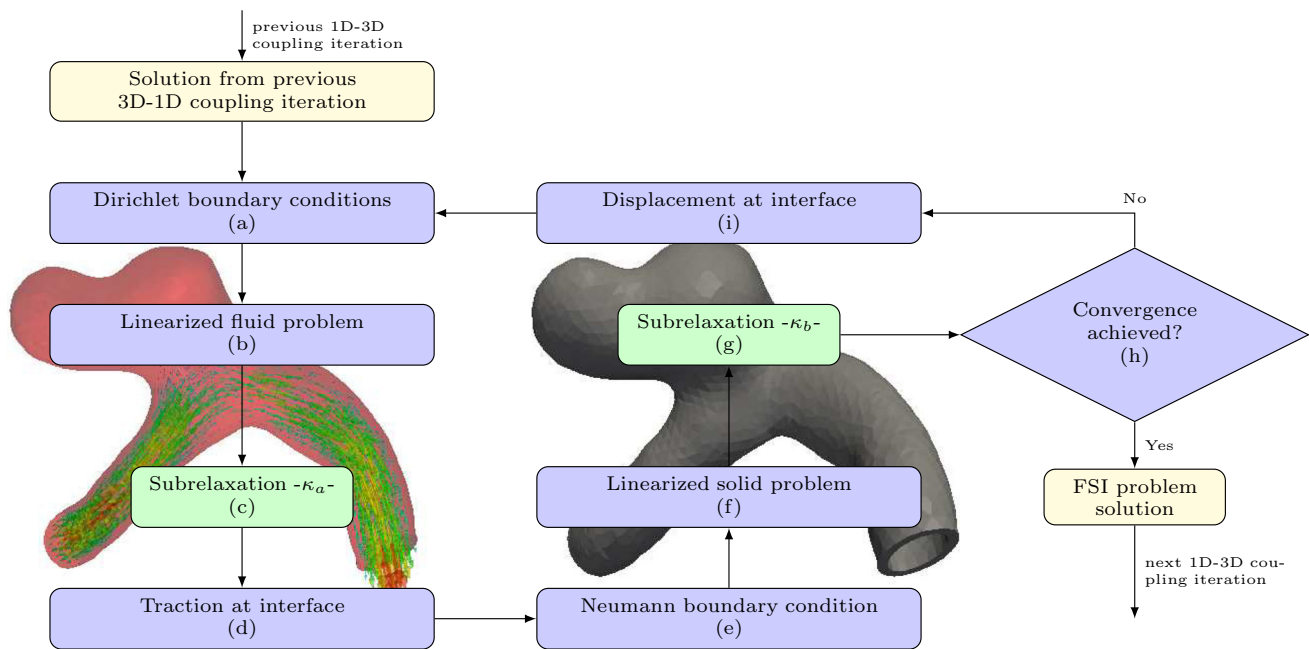


Fig. 3 Fluid–structure iterative coupling scheme

the entire CVS. Since there is no need for a full blood flow model of the complete circulation, dimensionally reduced models are instead employed to provide a fair description of the global dynamics which remains very accurate in terms of average quantities, such as average pressure and flow rate. Within this context, the entire CVS is modeled through different hemodynamic components which are properly coupled.

Concerning the so-called defective 3D problem, in the present work we have employed the variational setting developed in Blanco et al. (2007), in which the coupling of 3D and 1D models is understood and posed as a problem of coupling models with incompatible kinematics. This approach yields as a consequence that the traction field over the 3D interface is constant and points in the normal direction. This assumption is reasonable since we are assuming that the 3D model has been isolated from the rest of the arterial system at points where the velocity profile is fully developed and the Reynolds number is moderately high (maximum Reynolds number of approximately 1500). Model coupling is performed using a strong coupling approach based on the Broyden method, as described in Leiva et al. (2011).

Next, a brief description of the above-mentioned components is given:

Closed circulation component (CC) This component contains the arterial and venous systems, coupled through the cardiac circulation and the peripheral beds. Therefore, the cardiac ejection which inflows into the aortic root results from the interaction between the cardiac and systemic dynamics. The arterial network is modeled using the well-known 1D equations (which are omitted here for the sake of brevity)

considering a constitutive model for the arterial wall accounting for the elastic behavior of the elastin matrix and collagen fibers as well as the viscoelastic response of the structure. Such viscoelastic response is actually a composition of the viscoelasticity of the smooth muscle and that provided by surrounding tissues. It is important to note that the incorporation of this viscoelastic effect contributes to eliminate spurious reflections at coupling interfaces (Formaggia et al. 2013). The interested reader is directed to Blanco et al. (2007) for the detailed description of the equations for the 1D model, and to Blanco and Feijóo (2013) for the full description of the closed-loop circulation.

Specific-vessel component (SV) The arterial segment which demands a detailed description of the physical phenomena is represented using the model elaborated above in Sects. 2 and 3, that is: the three-dimensional behavior of the blood flow and its interaction with the arterial wall (see Sect. 3), considering the initially loaded—with pressure and tethering stresses—geometry (see Sect. 2). Eventually, coupling with surrounding tissues and residual stresses can be introduced in this component.

4.2 Strong coupling strategy

Consider that the closed circulation (CC) component is linked through C bonds to the specific-vessel (SV) component. For the sake of readability, we will call simply by “1D–0D” the CC component and by “3D” the SV component.

In Fig. 4, the schematic coupling is described. In this scheme, a single coupling point condenses the C coupling

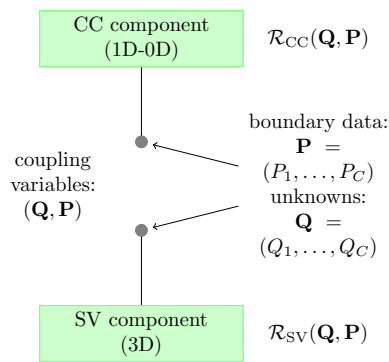


Fig. 4 Idealization of system as black-box component, coupling variables and chosen boundary data for components

interfaces. Therefore, at such coupling point, we have the unknown vector containing all the unknowns of the coupled system, namely $(\mathbf{Q}, \mathbf{P})$, being (Q_i, P_i) , $i = 1, \dots, C$ the pair of coupling variables at each coupling interface. In the present case, these variables stand for the flow rate and the normal component of the traction vector (considered constant over the 3D model boundary).

Hence, the coupling problem amounts to find the pair $(\mathbf{Q}, \mathbf{P})$ such that certain coupling equations are satisfied, that is

$$\begin{aligned} \mathcal{R}_{CC}(\mathbf{Q}, \mathbf{P}) &= 0, \\ \mathcal{R}_{SV}(\mathbf{Q}, \mathbf{P}) &= 0, \end{aligned} \quad (36)$$

where $\mathcal{R}_{CC}$ and $\mathcal{R}_{SV}$ are residuals defined accordingly. In the fluid flow problem, those residuals consist of equations of continuity of mass and continuity of the normal component of the traction vector (hereafter denoted simply by “coupling pressure”).

From the vector of unknowns $(\mathbf{Q}, \mathbf{P})$, it is now required to define those quantities that will turn into boundary data for each component within the iterative scheme. In this work, we choose the coupling pressure to be boundary data for both components. Hence, the system of coupling equations (36) results

$$\begin{aligned} \mathcal{R}_{CC}(\mathbf{Q}, \mathbf{P}) &= \mathbf{Q} - \mathcal{Q}_{CC}(\mathbf{P}) = 0, \\ \mathcal{R}_{SV}(\mathbf{Q}, \mathbf{P}) &= \mathbf{Q} - \mathcal{Q}_{SV}(\mathbf{P}) = 0, \end{aligned} \quad (37)$$

where $\mathcal{Q}_{CC}$ and $\mathcal{Q}_{SV}$ are compact vector notations for the operators related to the CC and SV components, respectively. These operators encompass the 1D–0D problem [see Blanco et al. (2013b)] and the 3D problem (see Sects. 2, 3) with $\mathbf{P} = (P_1, \dots, P_C)$ as boundary data at the C coupling interfaces. This system contains $2C$ equations and can be written as follows

$$\begin{aligned} Q_i - \mathcal{Q}_{CC}^i(P_1, \dots, P_C) &= 0 \quad i = 1, \dots, C, \\ Q_i - \mathcal{Q}_{SV}^i(P_1, \dots, P_C) &= 0 \quad i = 1, \dots, C. \end{aligned} \quad (38)$$

Thus, for instance, observe that $\mathcal{Q}_{SV}^i$ stands for the flow rate operator [see Eq. (35)] given by the SV component at i th coupling interface, after solving the FSI problem, for a given set $(P_1, \dots, P_C)$ of boundary data (see Sect. 3). Note that, according to Eq. (30), the forcing term for the fluid problem is given by the traction vectors acting at the boundaries $\partial \Upsilon_t^{A,i}$, $i = 1, \dots, C$. A constant normal traction field over those boundaries is imposed [as derived from Blanco et al. (2007), see discussion above]; thence it follows that in Eq. (30), we have $(t_t^1, \dots, t_t^C) = (P_1, \dots, P_C)$.

In the continuous setting, this system of nonlinear equations holds for every time. When approximating in time, system (38) is solved iteratively until convergence is achieved at each time step. The problem of solving (37) iteratively can be tackled using any suitable method for systems of nonlinear equations. Previous experiences dictate the use of a Broyden algorithm which avoids the computation of derivatives while preserving good convergence properties (Blanco et al. 2013b; Malossi et al. 2013; Leiva et al. 2011). In turn, at each coupling iteration (e.g., at each Broyden iteration), the entire FSI algorithm described in Fig. 3 is executed when solving the SV component. Thus, the iterations ruled by the coupling algorithm embrace the FSI iterations, while such coupling iterations are embraced by the temporal loop.

5 Complementary modeling issues

5.1 Constitutive modeling

We make use of the most extensively used constitutive model to represent the behavior of elastin and collagen components on the arterial wall, as proposed in Holzapfel and Gasser (2000).

In this constitutive framework, the tissue is modeled as a hyperelastic composed material in which the elastin matrix is taken into account through a Neo–Hookean isotropic term, and the collagen fibers are considered by an anisotropic term acting in two given directions. In the arterial wall, different layers can be distinguished, and for each layer, the same type of strain energy function is assumed, with a different set of material parameters and orientation characterizing the families of collagen fibers.

Hence, the strain energy function of the material takes the form:

$$\psi = \frac{c}{2}(I_1 - 3) + \frac{k_1}{2k_2} \sum_{i=4,6} \delta_i \left\{ e^{k_2(I_i - \lambda_i^0)^2} - 1 \right\} \quad (39)$$

where $\delta_i = 1$ if $I_i > \lambda_i^0$; $\delta_i = 0$ otherwise, and

$$I_1 = \text{tr}(\mathbf{C}_m), \quad I_i = \mathbf{a}_i \cdot (\mathbf{C}_m \mathbf{a}_i), \quad i = 4, 6, \quad (40)$$

with $\mathbf{C}_m = \mathbf{F}_m^T \mathbf{F}_m$, c is the material parameter which characterizes the stiffness of the elastin; the stiffness of collagen fibers is characterized by k_1 and k_2 , while the recruitment stretch is λ_i^0 , and the directions of orientation of fibers are $\mathbf{a}_i$, $i = 4, 6$, which are vectors defined in the material configuration Ω_m . The invariants I_4 , I_6 measure the stretch in the direction of the fibers as a consequence of the deformation state of the body. The exponential term will be active only if the stretch of the collagen fibers corresponds to a recruited state, that is $I_i > \lambda_i^0$, $i = 4, 6$. The passive response of the smooth muscle is taken into account in a compatible manner with the simplified 1D model through an independent ring model [see viscoelastic terms detailed in Blanco et al. (2010)].

5.2 Surrounding tissues

For the surrounding media, we assume that the effective behavior is viscoelastic. Then, the load $\mathbf{t}_s^E$ depends on the displacement and on the velocity of the solid wall at $\partial\Omega_s^E$, that is

$$\mathbf{t}_s^E = k_e (\mathbf{u}_s - \mathbf{u}_0) + k_v \frac{\partial \mathbf{u}_s}{\partial t} \Big|_{\partial\Omega_s^E}, \quad (41)$$

where k_e and k_v are effective elastic and viscoelastic parameters, respectively, and $\mathbf{u}_0$ defines a reference position for the elastic response.

5.3 Numerical approximation

The formulations described up to this point are general, and then independent from the numerical methods employed in the approximation of the problem. In this work, for both the fluid and the solid problems, the finite element method for the spatial discretization is used. The time integration is treated with an implicit Crank–Nicolson method. The finite element techniques employed regarding both solid and fluid problems are detailed next.

5.3.1 Numerical approximation for solid problems

The *forward* and *preload* problems require further treatment at the discrete level for improving preconditioning. This is accomplished by modifying the variational formulations introducing a consistent zero-order term on the linearized (forward and preload) problems. That is, the left-hand side of the first equation of (20) for the forward problem (component $a_s(\delta \mathbf{u}_s, \hat{\mathbf{u}}_s)$) includes the term

$$\int_{\Omega_s} \gamma \nabla_s \delta \mathbf{u}_s \cdot \nabla_s \hat{\mathbf{u}}_s \, d\Omega_s, \quad (42)$$

where γ can be regarded as a viscoelastic parameter. Note that this term is nullified when convergence is achieved, i.e., as $\delta \mathbf{u}_s \rightarrow 0$ (recall that $\delta \mathbf{u}_s = \mathbf{u}_s^{k+1} - \mathbf{u}_s^k$). Analogously, an identical term is added for the preload problem described by Eq. (66).

Linear finite elements for both displacement and pressure fields are used for the space discretization. To stabilize the problem in the sense of the inf-sup condition, the linearized (forward and preload) problems are modified adding a diffusive term in the pressure equation. That is, the left-hand side of the second equation of (20) for the forward problem includes the term

$$\int_{\Omega_s} \frac{h^2}{\nu} \nabla_s \delta \lambda_s \cdot \nabla_s \hat{q}_s \, d\Omega_s, \quad (43)$$

where h is a characteristic length of the finite element and ν a given stabilization parameter. Analogously, an identical term is added for the preload problem described by Eq. (66). This is a standard Galerkin–least squares stabilization-like term as proposed in Franca and Hughes (1988) and Hughes et al. (1989).

5.3.2 Numerical approximation for fluid problem

Equations (30)–(32) for the fluid problem are linearized using a standard fixed-point method as detailed in “Appendix 2”. As before, for the flow problem, the spatial discretization is carried out with tetrahedral elements with linear interpolation for velocity and pressure fields, and with the velocity field enriched using bubble functions to ensure stability in the sense of the inf-sup condition.

6 Numerical examples

6.1 Idealized common carotid artery

6.1.1 Problem description

A straight pipe representing an idealized human common carotid artery is studied. In the first place, the preload problem is solved to obtain the material configuration, assuming the initial—given—configuration has been extracted from medical images. Then, we proceed to solve the 3D FSI problem coupled with the 1D–0D model of the CVS.

Hereafter, we will refer to the fluid pressure field at a given time t as p_t , and define a baseline pressure level $p_{\text{base}} = 9 \cdot 10^4 \frac{\text{dyn}}{\text{cm}^2}$ (67.5 mmHg) and the relative pressure field at any given time as $p_t^{\text{rel}} = p_t - p_{\text{base}}$.

Six different cases are analyzed with two main objectives: (1) to assess the relevance of taking into account the complete load state at the so-called *initial* geometry, sep-

Table 2 Description of scenarios considered for the idealized carotid segment

Scenario	Material configuration	Preload problem loads		FSI and forward problem loads	
		Internal pressure	Tethering forces	Internal pressure	Tethering forces
(a)	Preload ^a	p_{base}	$\lambda_C = 1.2$	p_t	$\lambda_C = 1.2$
(b)	Preload ^a	p_{base}	$\lambda_C = 1.0$	p_t	$\lambda_C = 1.0$
(c)	Preload ^a	0	$\lambda_C = 1.2$	p_t^{rel}	$\lambda_C = 1.2$
(d)	Image ^b	–	–	p_t^{rel}	$\lambda_C = 1.0$
(e)	Image ^b	–	–	p_t	$\lambda_C = 1.2$
(f)	Image ^b	–	–	p_t	$\lambda_C = 1.0$

Loading conditions for the preload and FSI forward problems are presented

^a Indicates that the material configuration is known after solving the preload problem

^b Indicates that the material configuration is directly given by the geometry extracted from the image

arating the inner pressure load and the tethering forces; and (2) to analyze the pitfalls when considering alternative strategies to simulate the system response without having solved the preload problem. Thus, the following scenarios are considered:

- the preload problem is solved considering that the initial geometry is a configuration at mechanical equilibrium with a constant internal reference pressure p_{base} and that presents a known pre-stretch level $\lambda_C = 1.2$ and the FSI problem is solved from that configuration onwards,
- the preload problem is solved considering that the initial geometry is a configuration at mechanical equilibrium only with a constant internal reference pressure p_{base} , the length of the material configuration is equal to the initial length ($\lambda_C = 1.0$) and the FSI problem is solved from that configuration onwards,
- the preload problem is solved considering that the initial geometry presents a known pre-stretch level $\lambda_C = 1.2$ and no inner pressure is considered; the FSI problem is solved from that configuration onwards, assuming that the solid is equilibrated by a relative pressure p_t^{rel} ,
- no preload problem is solved and the constitutive equation is defined on the initial geometry, for the FSI problem is considered that the solid is a configuration at mechanical equilibrium with the defined relative pressure p_t^{rel} , i.e., neither the full level of pressure nor the stretch are considered in the simulation,
- no preload problem is solved and constitutive equations are defined on the initial geometry; the FSI problem is set in that configuration by firstly applying a known pre-stretch level $\lambda_C = 1.2$ as well as the full level of inner pressure p_t exerted by the blood flow at any given time t .
- no preload problem is solved and constitutive equations are defined on the initial geometry; the FSI problem is set in that configuration by applying the full level of inner pressure p_t exerted by the blood flow at any given time t and considering that the length of the vessel is

constant and equal to the corresponding for the initial state ($\lambda_C = 1$).

Note that: Case (a) corresponds to the most accurate scenario. Cases (b) and (c) are expected to give insight about the importance of taking into account the full load state when solving the preload problem (internal pressure and pre-stretch, respectively). Finally, cases (d), (e) and (f) will serve the purpose of comparing against standard approaches to blood flow and arterial wall simulations that completely neglect the preload problem whatsoever. Table 2 summarizes the loading conditions used to solve both the preload and the FSI problems for each test case.

For each case, we will denote as *diastolic baseline configuration* to the equilibrium geometry obtained when the vessel is loaded with the constant reference pressure $p = p_{\text{ref}}$ and the corresponding tethering forces. That means that for cases (a), (b), (c) and (d), the geometry will be that obtained from the image, while for cases (e) and (f), the geometry is considerably different due to loading conditions (see Table 4).

6.1.2 Geometrical and constitutive parameters

The idealized arterial segment is assumed to have been extracted from a medical image, with inner radius and thickness equal to $r_i = 0.37$ cm and $e = 0.063$ cm, respectively [data consistent with a common carotid artery vessel as given by Avolio (1980)], and the length of the pipe is $l = 8.9$ cm. The arterial wall consists of two layers, being the inner layer (media) two times the size of the external layer (adventitia).

The setting of material parameters is inspired in the data fitting presented in Zulliger et al. (2004) for a rat carotid artery, adapted according to the following criteria:

- adventitia layer mechanical properties are one order of magnitude smaller than the ones corresponding to the media layer,
- the inner radius of the equilibrium configuration when the vessel is loaded with a constant inner pressure level

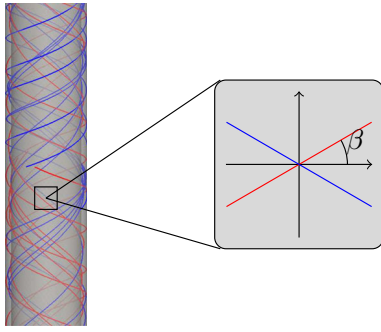


Fig. 5 Representation of the helicoidal arrangement of collagen fibers for the straight pipe. The orientation of the fibers is defined for media and adventitia layers through the angle β (see Table 3 for parameter values)

of $p = 1.6 \times 10^5 \frac{\text{dyn}}{\text{cm}^2}$ (120 mmHg) is approximately 5 % higher than the corresponding to a similar situation when the pressure level is at $p = 8 \times 10^4 \frac{\text{dyn}}{\text{cm}^2}$ (60 mmHg); these pressure levels correspond to reasonable systolic and diastolic pressure values, respectively,

- (3) two families of collagen fibers are considered with helicoidal structure in the circumferential direction (see Fig. 5), with a characteristic angle β for each family and material layer. The values of β for media and adventitia layers are taken from Holzapfel and Gasser (2000); the orientation of the fibers is in each case determined for the previously defined *diastolic baseline configuration*.

In addition, these collagen fibers are considered to be recruited when they are stretched beyond the *diastolic baseline configuration* Ω_{base} . Let $\mathbf{x}_{\text{base}}$ be the coordinates in this domain, and $\mathbf{u}^{\text{base}}$ be the displacements mapping this configuration with the material domain, with the associated deformation gradient $\mathbf{F}^{\text{base}} = \mathbf{I} + \nabla_{\mathbf{m}} \mathbf{u}^{\text{base}}$. Then for each scenario, the recruitment stretches λ_i^0 are defined via

$$\lambda_i^0 = \mathbf{a}_i \cdot (\mathbf{C}_m^{\text{base}} \mathbf{a}_i), \quad i = 4, 6, \quad (44)$$

with $\mathbf{C}_m^{\text{base}} = (\mathbf{F}_m^{\text{base}})^T \mathbf{F}_m^{\text{base}}$. To obtain a similar collagen behavior (within the physiological pressure range) to the presented in Zulliger et al. (2004) (where collagen is recruited at the zero-load configuration), the parameter k_2 [see Eq. (39)] controlling the exponential response of the fiber is increased. Table 3 presents the final material setting for this problem.

6.1.3 Discretization

The total number of nodes for the fluid domain is 31251. The fluid mesh is generated using a Delaunay method. The arterial wall is discretized using a semi-structured mesh of tetrahedral elements. This solid mesh was constructed through an extrusion of the surface representing the lumen boundary in

Table 3 Material properties considered in the numerical simulation of the carotid

Property	Media	Adventitia
Thickness (cm)	0.042	0.021
$c_{\text{elast}} \left(\frac{\text{dyn}}{\text{cm}^2} \right)$	8×10^6	8×10^5
$k_1 \left(\frac{\text{dyn}}{\text{cm}^2} \right)$	32×10^4	32×10^3
k_2	20.0	20.0
β	29.26°	65.37°

the normal direction. Prisms were generated from triangles, and then tetrahedra were constructed into the prisms. The resulting mesh consists of six layers of prisms in the radial direction (the first four inner element layers correspond to the media, and the two outer layers represent the adventitia). The total number of nodes for the solid domain is 94482. The time step in the simulation is taken $\Delta t = T/1000$, where T is the cardiac period.

6.1.4 Parameters for numerical simulation

Numerical parameters used are $\gamma = 0.01 \cdot c_{\text{elast}}$ for the zero-order preconditioning term [see (42)] and $\nu = 0.1 \cdot c_{\text{elast}}$ for the pressure stabilization [see (43)].

In the case of the preload problem, a subrelaxation step at each iteration of the Newton–Raphson scheme is introduced, and the value adopted for this subrelaxation step is 0.6. The solution is achieved via ten load steps in which the displacement at both ends of the cylinder (to define the pre-stretch, see Sect. 6.1.5) and the inner pressure (to define the preload pressure) are gradually incremented up to the desired values.

For the FSI problem, the chosen subrelaxation parameters are $\kappa_a = \kappa_b = 0.4$ (see Fig. 3; Sect. 3.4), while the parameter ι controlling the update of the Neumann boundary condition for the arterial wall at $\partial \Omega_s^W$ is defined by $n_s = 6$ (see Sect. 3.4).

6.1.5 Boundary conditions

Preload problem For the preload problem, Dirichlet boundary conditions are imposed to the displacement in the axial direction at non-physical boundaries, ensuring that a certain pre-stretch level is achieved. That is, for scenarios (a) and (c) an axial displacement of 0.7416 cm at each end of the cylinder is imposed, while for scenario (b), null axial displacements are considered. Neumann conditions are imposed at $\partial \Omega_s^W$ given by the constant p_{base} pressure load for scenarios (a) and (b), while in scenario (c), $\partial \Omega_s^W$ is set as a homogeneous Neumann boundary. In order to avoid rigid rotations, at each non-physical boundaries $\partial \Omega_s^{A,i}$, a penal-

ization over the tangential displacement is added to four diametrical nodes.

Finally, for the active force acting on the external boundary $\partial\Omega_s^E$ (see Sect. 5.2, Eq. (41)), the following assumptions are considered:

- the reference displacement for the elastic component is set at the *diastolic baseline configuration*, i.e., $\mathbf{u}_0 = \mathbf{u}_s^{\text{base}}$,
- the velocity of the body at the *baseline state* (relaxed diastolic state) is small.

Hence, the force given by external tissues is neglected in the preload problem, resulting in an homogeneous Neumann boundary condition over the corresponding boundary.

1D–3D FSI problem Boundary conditions for both the fluid and the solid at $\partial\Omega_s^W$ are taken as described in Sect. 3. In addition, coupling (Neumann-like) boundary conditions are set at non-physical boundaries for the fluid domain, providing a constant normal traction datum obtained from the 1D model through the dimensionally heterogeneous coupling algorithm. For the arterial wall, and analogously to the preload problem, Dirichlet conditions on the axial displacement are imposed at non-physical boundaries. Furthermore, Robin boundary conditions (at the time-discrete level) are incorporated on $\partial\Omega_s^E$ due to the action of the surrounding tissues (see Sect. 5.2, expression (41)). The parameters controlling the response of the external support are $k_e = 2 \times 10^2 \frac{\text{dyn}}{\text{cm}^2}$ and $k_v = 2 \times 10^3 \frac{\text{dyn}}{\text{cm}^2 \cdot s}$, while the reference for the elastic component is given by the displacements at the *diastolic baseline configuration*, i.e., $\mathbf{u}_0 = \mathbf{u}_s^{\text{base}}$.

6.1.6 Results

Figure 6 summarizes the results obtained for scenario (a); a comparison between the initial (image-based) and material geometries is displayed in Fig. 6a, b; the Von Mises stress state at the *baseline diastolic* and *systolic* configurations is shown in Fig. 6c, d, respectively; and, finally, the velocity profile representing the blood flow behavior is displayed in Fig. 6e. Since the flow rate contour (not shown here) is given by the 1D model and has a physiological shape, the blood velocity is within the physiological range.

Furthermore, Fig. 7 features the stress distribution for both diastole and systole states along thickness of the arterial wall. As a consequence of the tethering and internal pressure loads, the material presents a tensile state in both circumferential $\sigma_{\theta\theta}$ and axial σ_{zz} directions. The von Mises equivalent stress (σ_{vm}) is also displayed in the same figure. It is interesting to highlight that the behavior of the σ_{vm} differs from $\sigma_{\theta\theta}$ and σ_{zz} across the thickness of the arterial wall. While $\sigma_{\theta\theta}$ and σ_{zz} increase at systole in both layers, σ_{vm} slightly decreases in the inner layer and increases in the outer layer. The strong

discontinuity is related to the change in material properties from the media to the adventitia layers. All stresses are measured at a central slice of the vessel and mapped back to the image-based configuration.

In order to establish a comparison of the results for different scenarios, an overview of the predictions performed by the simulations is condensed in on Table 4 (geometrical landmarks of different configurations), Fig. 8 (velocity profiles at a given section of the pipe, located at $z = 0.67 \cdot l$, for $t = 0.35 \cdot T$), Fig. 9 (stress distribution along the thickness) and Table 5 (relative differences of the average stress at media and adventitia layers with respect to scenario (a)).

- Scenario (a). As said, this ideally represents the most accurate scenario. That is, the stress state (see Fig. 9) and the hydrodynamic quantities (see Fig. 8) can be considered as reference solution to discuss the differences caused by modeling simplifications in the rest of the scenarios considered (see items below). For instance, from Fig. 9 it is seen that the mechanical state of the arterial wall is pretty badly predicted by most of the scenarios, even changing the direction of stresses due to the lack of preload stresses.
- Scenario (b). From Fig. 9, it can be observed that, despite providing a good approximation of the geometry within the physiological range (see Table 4), neglecting the tethering forces given by the pre-stretch leads to major discrepancies when evaluating the stress state of the arterial wall tissue. As it is evident from the same figure, there are significant differences for both the circumferential and axial stresses and, as result, a severe underestimation of the von Mises equivalent stress. As displayed in Table 5, in average, the differences in the prediction of stresses reach 54, 104 and 66 % for $\sigma_{\theta\theta}$, σ_{zz} and σ_{vm} , respectively, with respect to scenario (a). As a consequence of the good estimation of the geometry achieved for this simple case, the blood flow behavior is fairly predicted, as can be observed in Fig. 8. Note that, compared with scenario (a), the shape of the profile has similar characteristics and the maximum velocity presents a difference of approximately 1.5 %.
- Scenario (c). Although the approximation of the geometry of the vessel is adequate within physiological range (see Table 4), the estimation of the stress state remains poor. Figure 9 clearly shows that the main flaw is in the prediction of the circumferential stress, since axial efforts are fairly represented. Since in this example the Von Mises stress invariant is primarily driven by the axial components, this scenario renders a good prediction for this quantity. Table 5 shows that the average stresses present differences up to 78, 12 and 14 % for $\sigma_{\theta\theta}$, σ_{zz} and σ_{vm} , respectively, with respect to scenario (a). Once

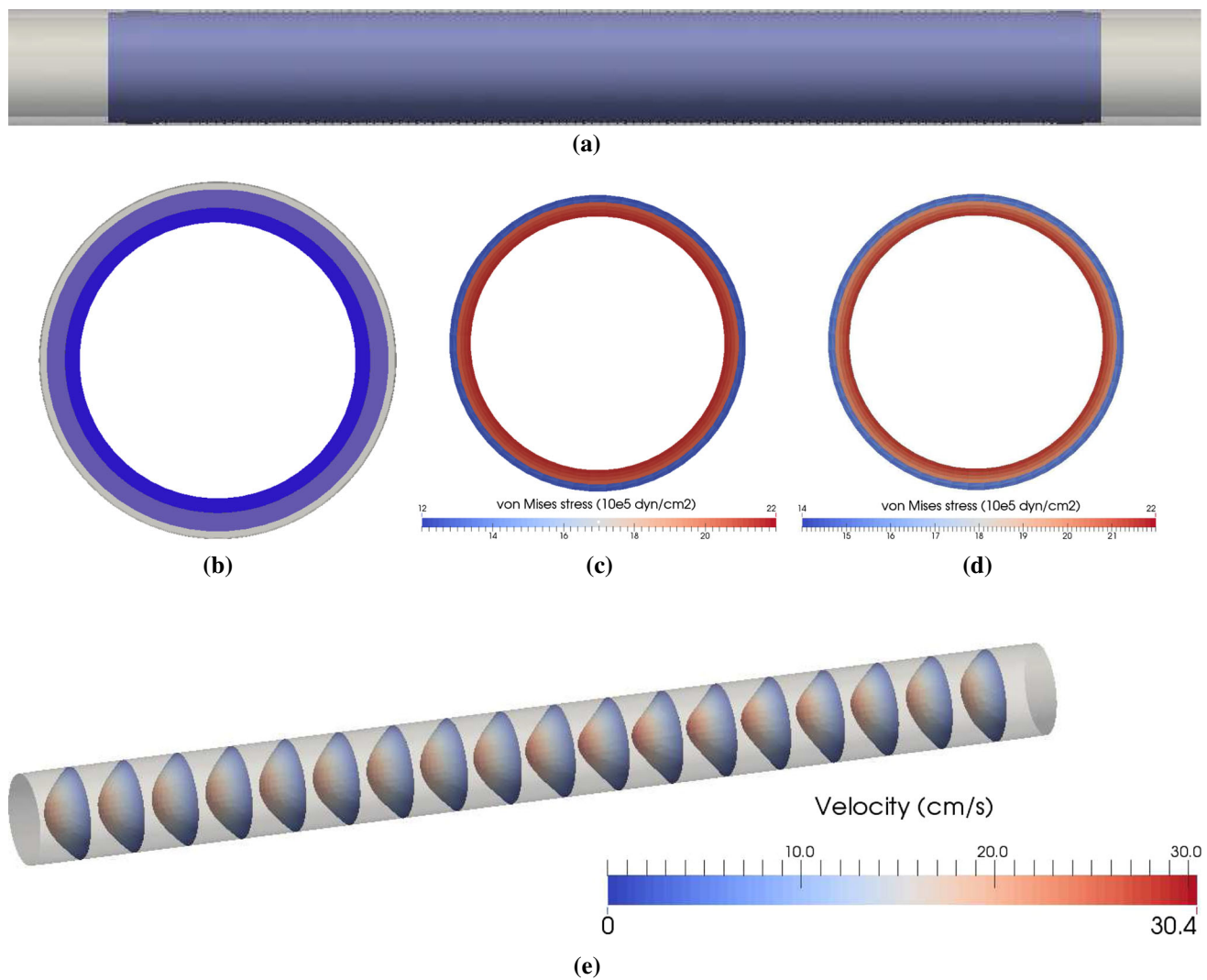


Fig. 6 Visualization of results for the 1D–3D FSI simulation of scenario (a). The notable difference on the stress levels is due to the material properties of the media and adventitia layers, **a** Comparison between *baseline diastolic* (blue) and material (translucent gray) configurations (side view), **b** comparison between *baseline diastolic* (blue) and mate-

rial (translucent gray) configurations (axial view), **c** Von Mises stress distribution on a central slice of the arterial wall at *baseline diastolic configuration*, **d** Von Mises stress distribution on a central slice of the arterial wall at systole, **e** velocity profile at systole ($t = 0.35 \cdot T$)

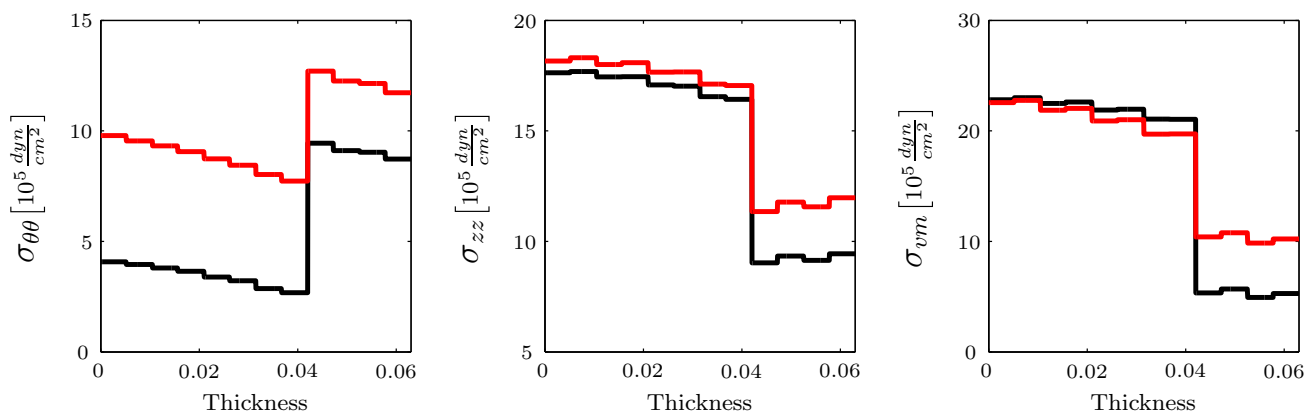


Fig. 7 Stress distribution along the thickness of the arterial wall for scenario (a) for diastole (black) and systole (red). Note the discontinuity introduced by the different material properties of the media and adventitia layers; also, that stresses are constant within each finite element

Table 4 Results overview displaying geometrical landmarks (inner radius r_i , length l and thickness t) of material, diastolic baseline and systolic configurations

Scenario	Geometrical landmarks							
	Zero-load			Diastolic ^a			Systolic ^b	
	r_i (cm)	l (cm)	t (cm)	r_i (cm)	l (cm)	t (cm)	r_i (cm)	t (cm)
(a)	0.352	7.42	0.088	0.370	8.90	0.063	0.383	0.060
(b)	0.349 (0.66)	8.90 (19.9)	0.065 (26.5)	0.370 (0.00)	8.90 (0.00)	0.063 (0.00)	0.382 (0.15)	0.060 (0.00)
(c)	0.371 (5.52)	7.42 (0.00)	0.073 (17.9)	0.370 (0.00)	8.90 (0.00)	0.063 (0.00)	0.383 (0.00)	0.060 (0.00)
(d)	0.370 (5.22)	8.90 (19.9)	0.063 (28.7)	0.370 (0.00)	8.90 (0.00)	0.063 (0.00)	0.384 (0.33)	0.061 (0.99)
(e)	0.370 (5.22)	8.90 (19.9)	0.063 (28.7)	0.387 (4.49)	10.89 (20.0)	0.049 (22.7)	0.396 (3.52)	0.051 (16.2)
(f)	0.370 (5.22)	8.90 (19.9)	0.063 (28.7)	0.394 (6.48)	8.90 (00.0)	0.061 (2.86)	0.405 (5.64)	0.059 (1.83)

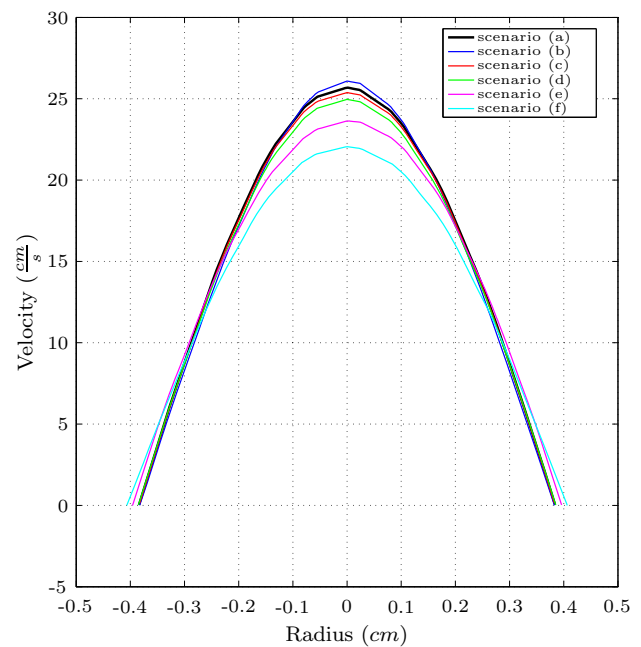
Measures taken at a central slice of the vessel. In parenthesis, percent differences with respect to scenario (a) are reported

^a Diastolic baseline configuration

^b Configuration at systole ($t = 0.35 \cdot T$)

again, similarly to the findings corresponding to scenario (b), the results for the blood flow behavior are in agreement with those obtained from scenario (a), as shown in Fig. 8.

- Scenario (d). The geometrical landmarks as well as the hydrodynamics are well predicted by this scenario, as observed in Table 4 and Fig. 8, respectively. As expected, the stress state predicted in this case is substantially different from that obtained in scenario (a). In fact, average relative differences of 81, 102 and 86 % are obtained for $\sigma_{\theta\theta}$, σ_{zz} and σ_{vm} , respectively (see Table 5). Recall that, due to the hypothesis considered for this case, stresses at the *baseline diastolic configuration* are null. For this case, it is interesting to contrast the differences of the stresses between diastole and systole (from here on referred to as $\Delta\sigma$). This comparison is presented in Fig. 10 and Table 6, where it can be appreciated that this scenario is also unable to predict $\Delta\sigma$ accurately, with relative differences up to 25, 181 and 27 % for $\Delta\sigma_{\theta\theta}$, $\Delta\sigma_{zz}$ and $\Delta\sigma_{vm}$, respectively.
- Scenario (e). Differences in geometrical landmarks are significant as observed in Table 4, with discrepancies of approximately 4 % in the internal radius for both the diastolic and systolic domains (leading to a disparity of roughly 10 % in the lumen area). Also note that the assumptions taken for this case lead to misrepresentation of the tissue thickness, which significantly impacts the mechanical response of the wall. Although similar flow rates are obtained between the current case and scenario (a), the increased cross-sectional area reduces the blood velocity as observed in Fig. 8, where it is appreciated a decrease of 8 % on the maximum velocity. Regarding the stress state, an adequate pattern for both circumferential and axial stresses is predicted, but they are overestimated when compared with scenario (a), as noticed in Fig. 9. Table 5 displays that relative differences reach 33, 34 and 33 % for $\sigma_{\theta\theta}$, σ_{zz} and σ_{vm} , respectively.

**Fig. 8** Velocity profile for different scenarios at time $t = 0.35 \cdot T$ measured over a cross-sectional line

- Scenario (f). Significant differences in the geometrical landmarks are obtained for both the diastolic and systolic domains, i.e., discrepancies of approximately 6 and 13 % are reported for the internal radius and lumen area, respectively. Similarly to scenario (e), due to the increased cross-sectional area, the blood velocity is reduced as shown in Fig. 8. This time the maximum velocity is underestimated, presenting a discrepancy of 14.1 compared with our reference scenario. Regarding the stress results, it is evident from Fig. 9 that this approach is completely unable to characterize the mechanical response of the tissue, since in average, the differences in the prediction of stresses reach 50, 103 and

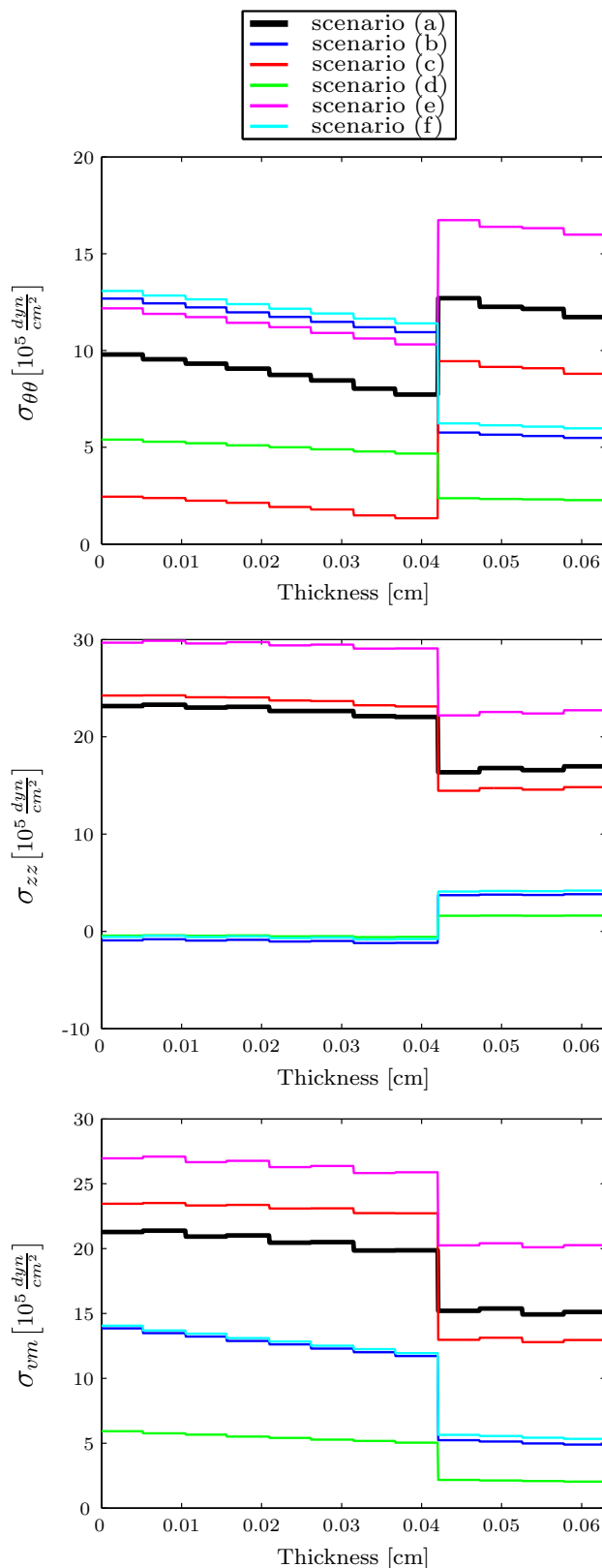


Fig. 9 Distribution of circumferential (*top*), axial (*middle*) and Von Mises (*bottom*) stresses at systole along the thickness of the arterial wall for the proposed scenarios [(a) black, (b) blue, (c) red, (d) green, (e) magenta, (f) cyan]

Table 5 Percentual differences with respect to scenario (a) of the average circumferential, axial and von Mises stresses for each material layer and proposed scenario. Measures taken on a central slice of the vessel at $t = 0.35 \cdot T$

Scen.	Discrepancies of average stress values (%)					
	Media			Adventitia		
	$\sigma_{\theta\theta}$	σ_{zz}	σ_{vm}	$\sigma_{\theta\theta}$	σ_{zz}	σ_{vm}
(b)	34.0	104	38.3	53.8	77.5	66.5
(c)	77.7	4.59	12.1	25.4	12.1	14.4
(d)	42.9	102	73.5	80.9	90.3	86.1
(e)	27.8	29.6	28.2	33.9	34.8	33.6
(f)	38.8	102.8	37.2	49.8	75.3	63.7

64 % for $\sigma_{\theta\theta}$, σ_{zz} and σ_{vm} , respectively, with respect to scenario (a).

From the discussion above, it can be inferred that no alternative scenario to case (a) is suitable to accurately evaluate the stress state of the arterial wall. It has been shown that it is necessary to solve the preload problem taking into account both the internal pressure and the axial pre-stretch. Moreover, it has been clearly demonstrated even in this simple geometrical model that the tethering forces, usually disregarded, are, as well as the inner pressure load, of the utmost importance for the determination of the stress state of the arterial tissue (Fig. 9 and Table 5).

For this particular example, due to the simplicity of geometry and the low compliance of the vessel, all scenarios predict comparable variations of the lumen radius. Consequently, the hydrodynamic variables feature similar behavior in all scenarios. This may not be the case for more compliant vessels or more tortuous domains, e.g., for the entire aorta or for some vessels containing aneurysms (Hsu and Bazilevs 2011), among others.

6.2 Idealized common carotid bifurcation

6.2.1 Problem description

This numerical example consists in a standard carotid bifurcation, with the fluid domain defined as in Bharadvaj et al. (1982). As before, the preload problem is solved to obtain the material configuration assuming that the initial geometry has been extracted from medical images. Then, the 1D–3D FSI coupled problem is solved. Once again, the baseline pressure level is defined $p_{\text{base}} = 9 \cdot 10^4 \frac{\text{dyn}}{\text{cm}^2}$ (67.5 mmHg), and given the pressure field p_t at the time t , the relative pressure level is $p_t^{\text{rel}} = p_t - p_{\text{base}}$.

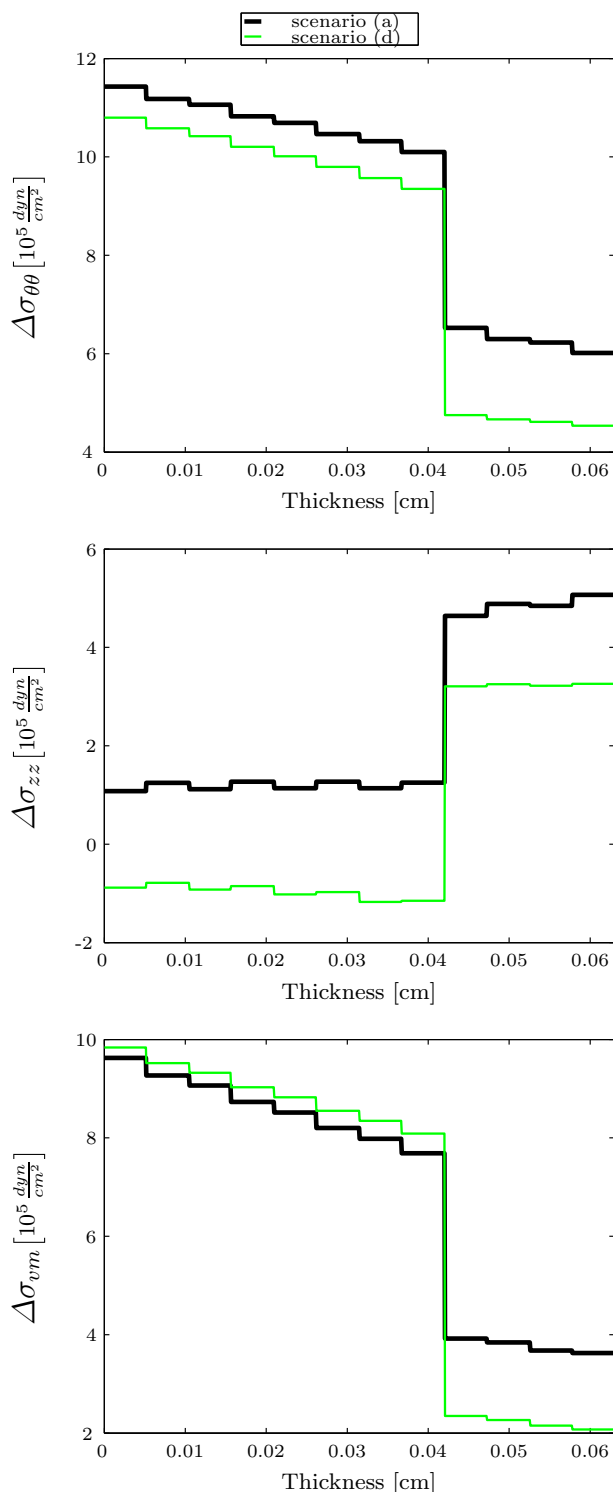


Fig. 10 Difference of stresses (circumferential, axial and Von Mises) between systole and diastole for scenarios (a) and (d) (black and green respectively)

In this case, we consider the two following scenarios:

- (a) the preload problem is solved considering that the initial geometry is a configuration at mechanical equilibrium with a constant internal reference pressure p_{base} and is

Table 6 Percentual discrepancies with respect to scenario (a) of the difference between systolic ($t = 0.35 \cdot T$) and baseline diastolic stresses ($\Delta \Sigma$) for each material layer. Measures taken on a central slice of the vessel

Scenario (d) : discrepancies of average $\Delta \sigma$ (%)					
Media			Adventitia		
$\Delta \sigma_{\theta\theta}$	$\Delta \sigma_{zz}$	$\Delta \sigma_{vm}$	$\Delta \sigma_{\theta\theta}$	$\Delta \sigma_{zz}$	$\Delta \sigma_{vm}$
6.19	181	2.87	25.8	33.6	26.8

also subjected to a known pre-stretch level $\lambda_C = 1.2$. Note that the FSI problem is solved from that configuration onwards,

- (b) the preload problem is solved considering that the initial geometry is a configuration at mechanical equilibrium with a constant internal reference pressure p_{base} and that is not pre-stretched ($\lambda_C = 1.0$); the FSI problem is solved from that configuration onwards, assuming that the solid is equilibrated by the full level of inner pressure p_t .

Note that, as before, case (a) constitutes the most realistic scenario and case (b) will serve the purpose of comparing with an alternative approach and to further quantify the influence of the tethering forces on the response of the arterial wall.

6.2.2 Constitutive parameters

Base values for the constitutive parameters employed in this example are identical to those presented for the straight pipe. As pointed out in Steinman et al. (2002), the carotid bulb and bifurcation zone can be subjected to elevated thickness values. In accordance with this, we have smoothly increased the thickness toward the bifurcation region up to two times the designated values at the inlet and outlet boundaries; this feature can be appreciated on Fig. 11.

Collagen orientation is given at the initial (image-based) configuration Ω^I . Moreover, the recruitment stretch for the fibers is defined from this configuration in an analogous manner to the previous example (in that case for Ω^{base}). Note that, for both cases, the image-based configuration is equivalent to the baseline diastolic domain.

6.2.3 Discretization

The total number of nodes for the discretization of the fluid domain is 27673. The arterial wall is discretized using the same strategy described in Sect. 6.1.3. The total number of nodes for the solid domain is 87751. The time step in the simulation is taken $\Delta t = T/1000$, where T is the cardiac period.

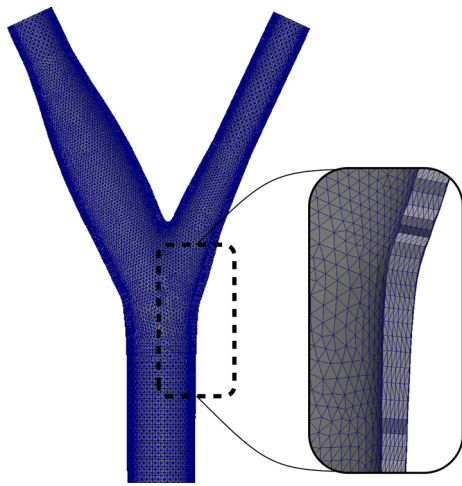


Fig. 11 Clipping of the arterial wall mesh through the symmetry plane

6.2.4 Boundary conditions

Preload problem For the preload problem, Dirichlet boundary conditions are imposed to the displacement in the axial direction at the non-physical boundary corresponding to the inlet, ensuring that a certain pre-stretch level is achieved. That is, for scenario (a) an axial displacement of 1.05 cm is considered, while for case (b), null axial displacement is imposed. Note that for case (a), the pre-stretch level is not strictly 20 %, but the magnitude of the imposed displacements represents

the 20 % of the length of the vessel projected to the axis of the common carotid (z cartesian coordinate). Neumann conditions are imposed at $\partial\Omega_s^W$ given by the constant p_{base} pressure load for both scenarios. For the external boundary $\partial\Omega_s^E$, homogeneous Neumann conditions for each case are considered based on the assumptions exposed in Sect. 6.1.5. Finally, at each distal boundary, a penalization over the mean displacement of the surface $\partial\Omega_s^{A,i}$ is considered, thus fixating the center of the outflow area.

1D–3D FSI problem Boundary conditions are analogous to those presented in Sect. 6.1.5.

6.2.5 Parameters for numerical simulation

The parameter setup for the numerical simulation is identical to the presented for the straight pipe problem (see Sect. 6.1.4).

6.2.6 Results: computation of material configuration and 1D–3D FSI simulation

For both cases, the baseline diastolic configuration is given and the preload problem solved in order to obtain the material configuration. Figure 12b, c show the resulting zero-load reference configurations for scenarios (a) and (b), respectively. From these images, the great influence of the pre-stretch on the computation of the material domain is clearly noticeable.

Figure 13 summarizes the obtained results for both scenarios regarding the mechanical response of the wall tissue, showing the von Mises equivalent stress distribution for the

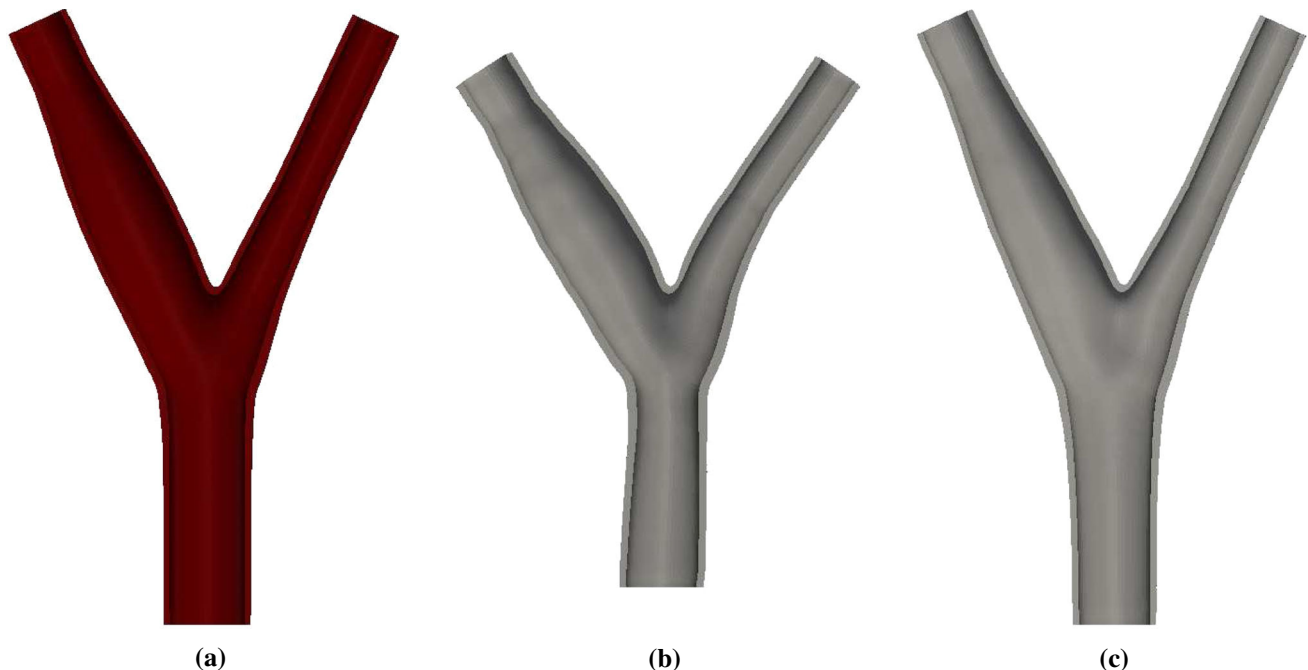


Fig. 12 Computation of material configurations, comparison between scenarios (a) and (b). **a** Baseline diastolic configuration, **b** computed zero-load reference configuration for scenario (a), **c** computed zero-load reference configuration for scenario (b)

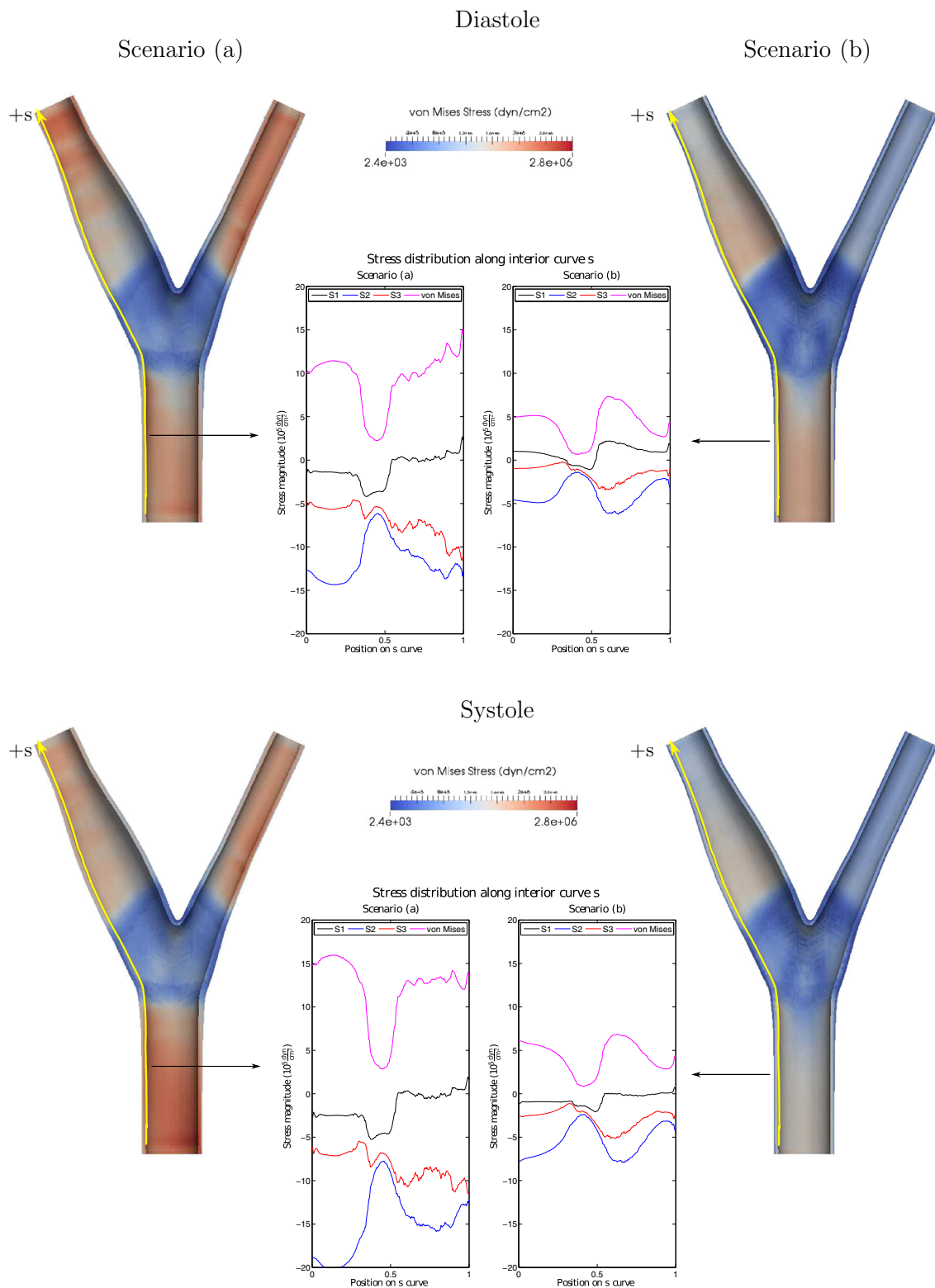


Fig. 13 Visualization of the computed stresses for scenarios (a) (left) and (b) (right) mapped into the image-based configuration. Figure displays the von Mises equivalent stress of the arterial wall tissue clipping through the symmetry plane of the geometry for $t = 0.235 \cdot T$ (in the context of this figure denoted as systole) and for the baseline diastolic

configuration (in the context of this figure denoted as diastole). Charts present the stress distribution (principal stresses in blue (S1), black (S2) and red (S3); von Mises equivalent stress in magenta) over an interior line s going through the common and internal carotid arteries

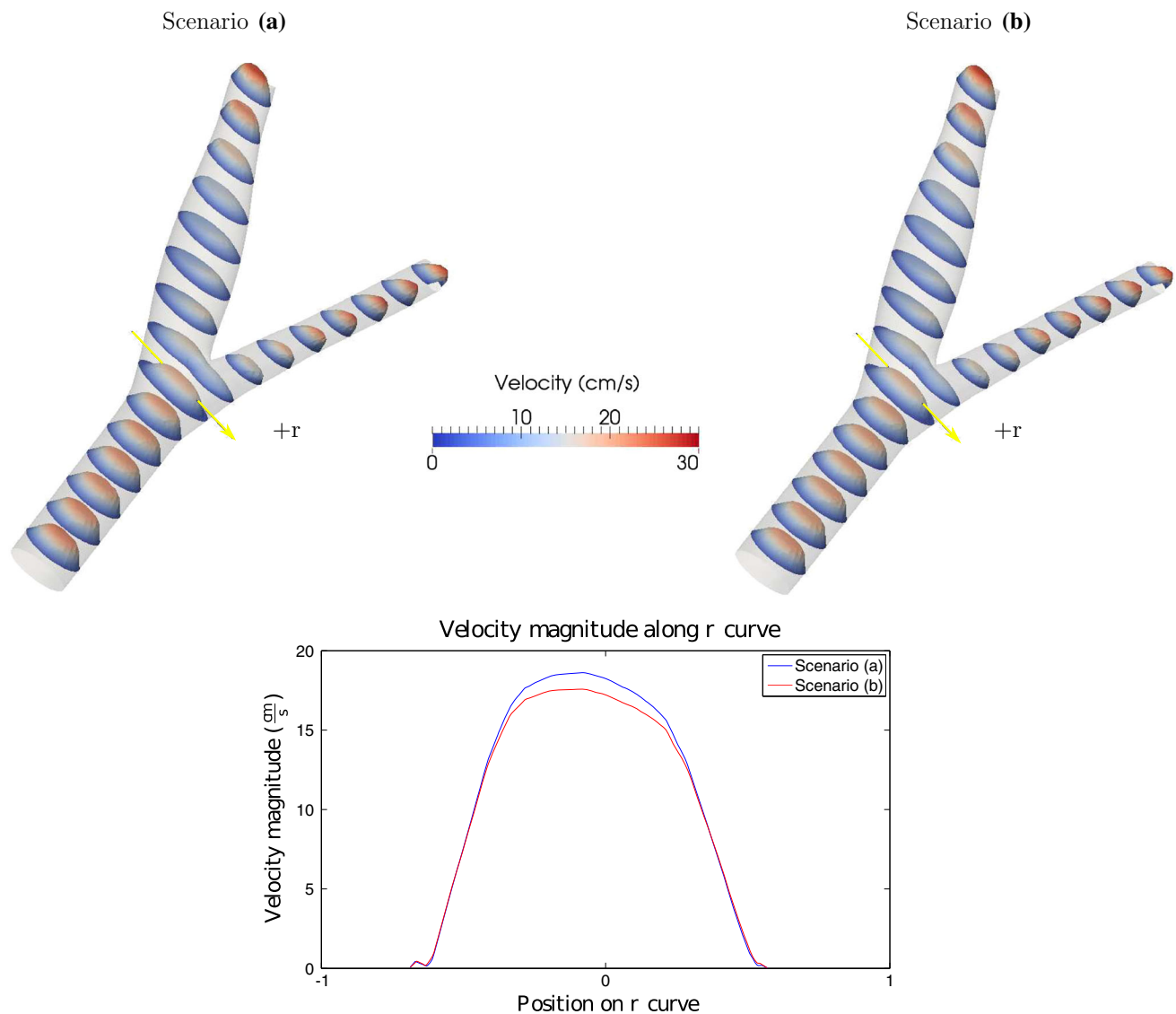


Fig. 14 Velocity profile for scenarios (a) and (b) at $t = 0.235 \cdot T$

carotid bifurcation at a given instant during the cardiac systole $t = 0.235 \cdot T$ and for the baseline diastolic configuration (result given by the preload problem). Moreover, the values of the principal stresses (S_1 , S_2 , S_3) over a curve s going through the interior of the vessel from the inflow boundary to one of the outlets (corresponding to the internal carotid) are detailed on the featured charts. Once again, it can be clearly appreciated that disregarding the tethering forces leads to an unrealistic evaluation of the stress state. Analyzing the principal stresses along the s curve at systole, the relative discrepancies with respect to scenario (a) report values up to 66 %, while for the von Mises stresses, the relative discrepancies reach 78 %.

The blood flow behavior is represented in Fig. 14, presenting the velocity profile on the deformed domain at different slices of the geometry and a chart comparing the veloc-

ity magnitude along a line through the bifurcation point at $t = 0.235 \cdot T$. It is observed that the velocity field shows a good agreement between both cases in all the domain; in the particular case of the velocity over the selected curve, consistency on the shape of the profile and a difference of approximately 5 % on the maximum reported value is shown. The same behavior is observed throughout the entire cycle (results not shown).

In this example, the relevance of accounting for the pre-stretch state of the arterial wall is further emphasized, evidencing the influence of the tethering forces on the mechanical response of the tissue. As previously noted in the straight pipe example, due to the low compliance of the vessel and the characteristics of this geometry, scenario (b) shows similar characteristics regarding the blood flow behavior.

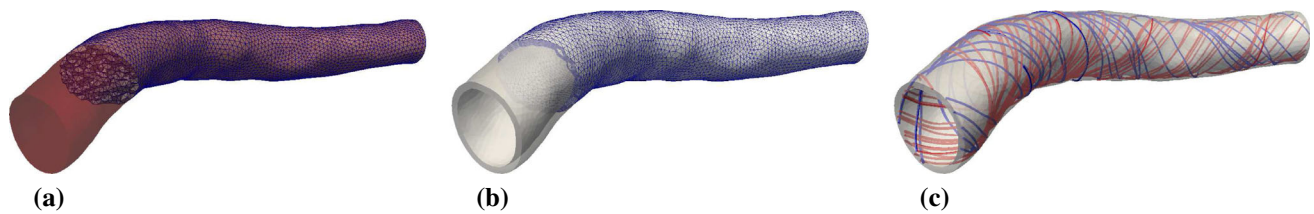


Fig. 15 Image-based fluid and solid domains, discretization and collagen orientation. **a** Initial image-based blood flow domain and discretization, **b** initial image-based (*baseline*) wall configuration and discretization, **c** collagen orientation setting for arterial wall

6.3 Patient-specific common carotid artery

6.3.1 Problem description

This numerical example consists in a patient-specific left common carotid artery extracted from a medical imaging study (computed tomography angiography). The inner radius of the segment varies between 0.25 and 0.33 cm, and its length is $l = 5.75$ cm measured over the centerline of the vessel. Both the blood flow domain and the arterial wall domain are presented in Fig. 15a and b, respectively. Analogously to the previous examples, two scenarios are analyzed:

- the preload problem is solved considering that the initial geometry is a configuration at mechanical equilibrium with a constant internal reference pressure p_{base} and is also subjected to pre-stretching. Recall that the FSI problem is solved from that configuration onwards,
- the preload problem is solved considering that the initial geometry is a configuration at mechanical equilibrium with a constant internal reference pressure p_{base} and that the position of the non-physical boundaries is fixed; the FSI problem is solved from that configuration onwards, assuming that the solid is equilibrated by the full level of inner pressure p_t .

For this example, the pre-stretch in scenario (a) is introduced by setting axial tethering forces at non-physical boundaries such that a certain prescribed displacement is achieved. Through this procedure, a shortening of approximately 7 % ($\lambda_C \approx 1.07$) is achieved. Also, it is important to remark that fixating the position of these boundaries in scenario (b) does not imply that the length of the segment is constant; in fact, a subtle shortening is observed when solving the preload problem because of the geometrical ability of the vessel to accommodate the curvature, obtaining a pre-stretch of approximately 1.5 % ($\lambda_C \approx 1.015$). As before, the baseline pressure level is defined $p_{\text{base}} = 9 \cdot 10^4 \frac{\text{dyn}}{\text{cm}^2}$ (67.5 mmHg).

6.3.2 Constitutive parameters

Base values for the constitutive parameters employed in this example are identical to those presented for the straight pipe (see Sect. 6.1). The thickness of the wall is defined based

on the maximum radius observed in the carotid segment, and taking into account the ratio of the inner radius to the thickness presented in Sect. 6.1, that is, $\frac{r_i}{e} \frac{0.37}{0.063}$. Hence, the resulting thickness for this test is $e = 0.056$ cm. The setting of the collagen fibers is presented in Fig. 15c.

6.3.3 Discretization

The total number of nodes for the discretization of the fluid domain is 30760. The arterial wall is discretized using the same strategy described in Sect. 6.1.3. The total number of nodes for the solid domain is 34465. The time step in the simulation is taken $\Delta t = T/1000$, where T is the cardiac period.

6.3.4 Boundary conditions

Preload problem For scenario (a), tethering forces in the normal direction are introduced through a penalization, forcing a displacement of the center of both non-physical boundaries equal to 0.2 cm. As stated before, these tethering forces induce a pre-stretch level of $\lambda_C = 1.07$. For scenario (b), the position of the center of the non-physical boundary is fixed through the same penalization method. Due to the geometrical characteristics of the vessel and the action of the internal pressure, the material configuration results to be shorter than the image-based initial domain, presenting a subtle pre-stretch level of approximately 1.5 % ($\lambda_C \approx 1.015$). Neumann conditions are imposed at $\partial\Omega_s^W$ given by the constant p_{base} pressure load for both scenarios. For the external boundary $\partial\Omega_s^E$, homogeneous Neumann conditions for each case are considered based on the assumptions exposed in Sect. 6.1.5.

1D–3D FSI problem Boundary conditions are analogous to those presented in Sect. 6.1.5.

6.3.5 Parameters for numerical simulation

The parameter setup for the numerical simulation is identical to the presented for the straight pipe problem (see Sect. 6.1.4).

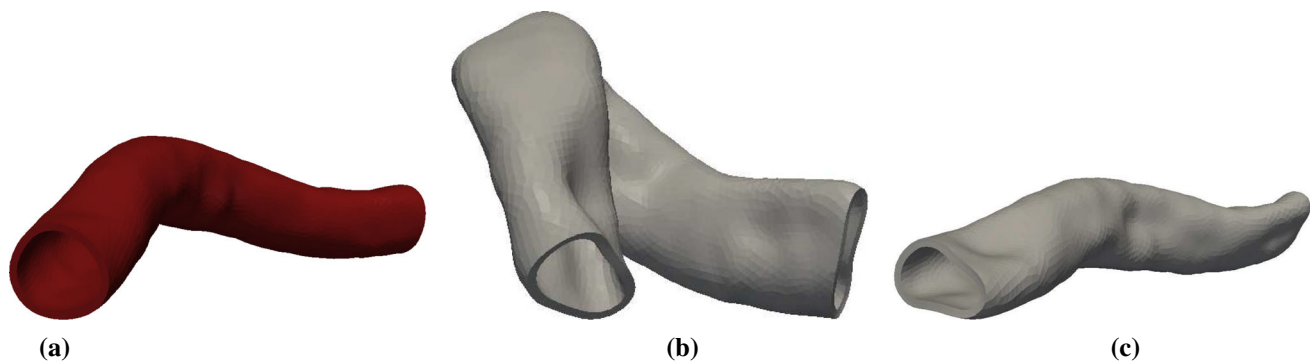


Fig. 16 Computation of material configurations, comparison between scenarios (a) and (b). **a** Baseline diastolic configuration, **b** computed zero-load reference configuration for scenario (a), **c** computed zero-load reference configuration for scenario (b)

6.3.6 Results: computation of material configuration and 1D–3D FSI simulation

As presented for the previous example (see Sect. 6.2), in this new problem the baseline diastolic configuration is given and the preload problem is solved in order to obtain the material configuration for the scenarios described above. The initial image-based (baseline) configuration is presented in Fig. 16a, while Fig. 16b, c displays the resulting zero-load reference configurations for scenarios (a) and (b), respectively. Again, from these images, the great influence of the different hypotheses regarding the tethering forces on the computation of the material domain is evident.

Figure 17 summarizes the results obtained for both scenarios regarding the mechanical response of the tissue wall. Particularly, the Von Mises equivalent stress σ_{vm} at the baseline configuration for scenarios (a) and (b) is presented on Fig. 17a and b, respectively. In turn, Fig. 17c, d displays the corresponding results for a systolic time instant ($t = 0.35 T$). The above-mentioned images feature, for every case, the stress state at the interior surface of the vessel and the transmural distribution along a longitudinal cut (left). Furthermore, areas featuring large stresses are highlighted in a threshold view, where only elements with $\sigma_{vm} \geq 2 \times 10^6 \frac{\text{dyn}}{\text{cm}^2}$ are shown (right).

These results show the impact of the different hypotheses assumed for each scenario on a realistic geometry. The vessel's curvature leads to different localizations (and values) of stress augmentation in addition to the previously reported differences in the general levels of stress. Moreover, Fig. 18 presents the relative discrepancy between the Von Mises stress values in the baseline and systolic ($t = 0.35 T$) states (reference stress values of $1 \times 10^6 \frac{\text{dyn}}{\text{cm}^2}$ and $2 \times 10^6 \frac{\text{dyn}}{\text{cm}^2}$ for each case), evidencing differences up to 270 %.

The blood flow behavior is shown in Fig. 19. Blood velocity profile in the deformed domain at different slices of the geometry is shown for two characteristic time instants: at systole $t = 0.35 T$ and at diastole $t = T$. As reported for the

previous numerical examples, it is observed that the velocity field shows small sensitivity to the stress state of the arterial wall.

7 Discussion and concluding remarks

7.1 Hemodynamic insight

The examples analyzed in the previous section shed light on the sensitivity of the arterial wall stress state and blood flow to the definition of the domain of analysis and loads considered to be applied in the image-derived configuration.

On one hand, these examples demonstrate that blood flow phenomena are not sensitive to the definition of the preload problem. That is, considering the image configuration as material configuration, or considering the true material configuration obtained after solving the preload problem (whatever the loads are), does not significantly affect the flow patterns. Although some debate may exist in the case of more compliant vessels and different flow regimes, this is reasonable for the cases analyzed within the present work. Similar results in this direction support this fact, as shown in Dempere-Marco et al. (2006) in the context of blood flow simulation comparing outcomes between compliant and rigid domains.

On the other hand, the reported results show that solving the preload problem (scenario (a)) including pressure and tethering loads is essential to correctly characterize the mechanical state of the arterial wall. In fact, the use of simplifying hypotheses irremediably leads to large deviations from the stresses computed in scenario (a).

Moreover, the obtained results correspond to the most realistic scenario through the consideration of fluid–structure interaction and coupling with a one-dimensional network of the rest of the arterial tree. This ensures a physiologically consistent hemodynamic environment for the region of interest.

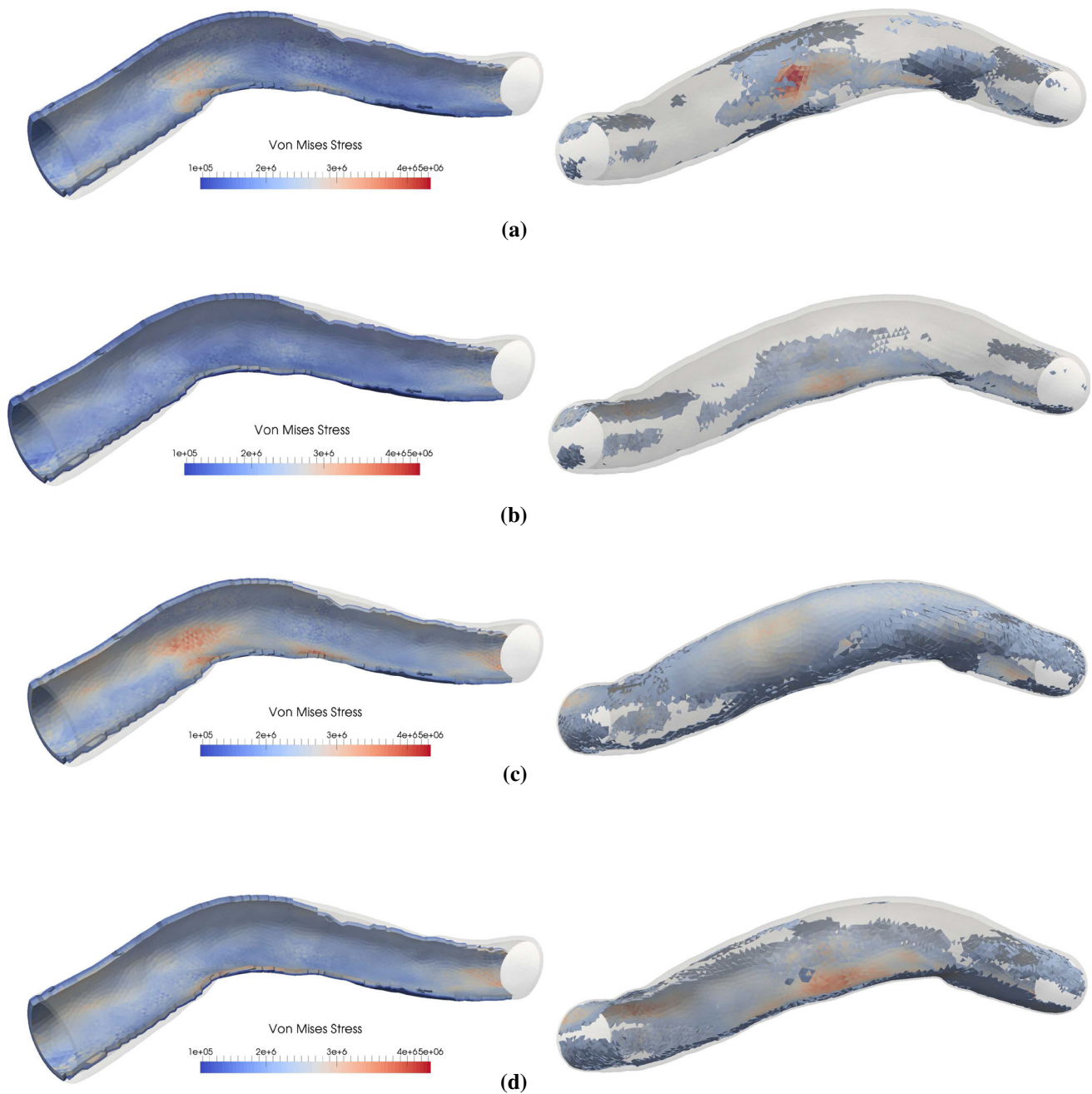


Fig. 17 Stress state at the arterial wall for scenarios (a) and (b) for the baseline diastolic and systolic configurations. **a** Scenario (a): Von Mises equivalent stress for baseline diastolic configuration, interior wall view (left) and threshold displaying most stressed regions ($\sigma_{vm} \geq 2 \times 10^6 \frac{\text{dyn}}{\text{cm}^2}$) (right). **b** Scenario (b): Von Mises equivalent stress for baseline diastolic configuration, interior wall view (left) and

threshold displaying most stressed regions ($\sigma_{vm} \geq 2 \times 10^6 \frac{\text{dyn}}{\text{cm}^2}$) (right). **c** Scenario (a): Von Mises equivalent stress for systolic configuration, interior wall view (left) and threshold displaying most stressed regions ($\sigma_{vm} \geq 2 \times 10^6 \frac{\text{dyn}}{\text{cm}^2}$) (right). **d** Scenario (b): Von Mises equivalent stress for systolic configuration, interior wall view (left) and threshold displaying most stressed regions ($\sigma_{vm} \geq 2 \times 10^6 \frac{\text{dyn}}{\text{cm}^2}$) (right)

7.2 Limitations

Most of the limitations of the present approach are shared with standard FSI approaches. As for the detailed 3D model, the constitutive parameters of the arterial wall and the definition of the arterial thickness are classical examples.

In turn, little is known about the interaction of the arterial wall with the surrounding media, and much is still to be done. The proposed approach relies on few works available in the literature, and this remains an open problem for the community. In addition, as commented in the Introduction, the origin of residual stresses due to growth and remodeling



Fig. 18 Relative discrepancy of Von Mises stresses between scenarios (a) and (b) for baseline (*left*) and systolic (*right*) configurations. The reference stress level have been defined as $\sigma_{ref} = 1 \times 10^6$ and $\sigma_{ref} = 2 \times 10^6$, respectively

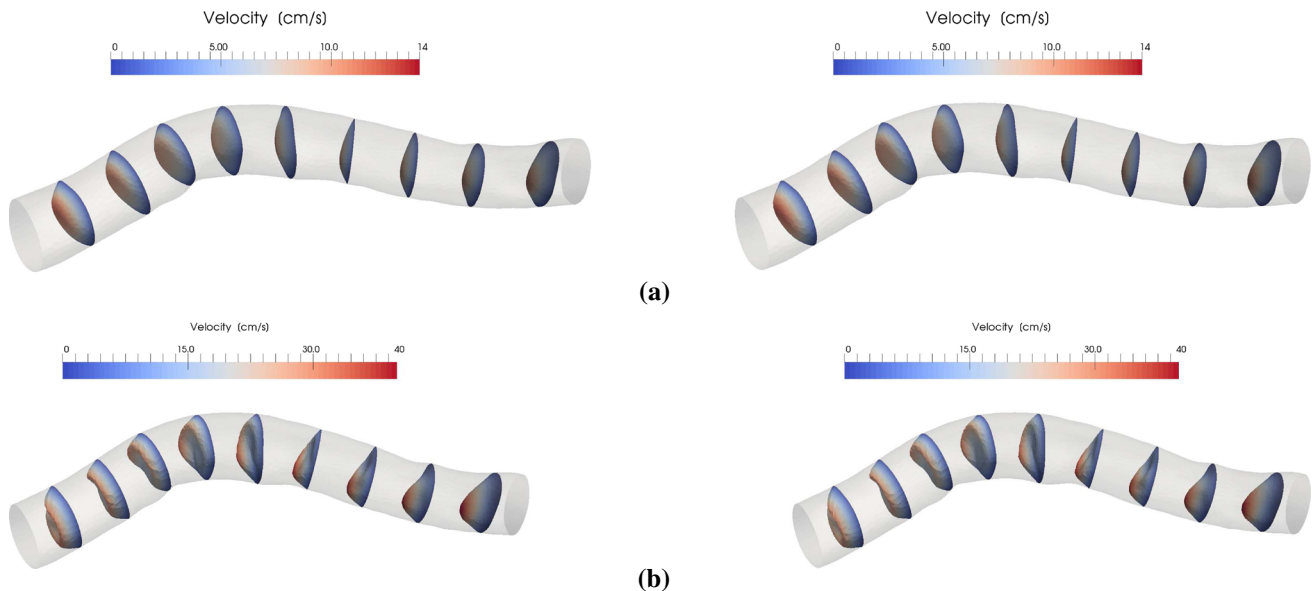


Fig. 19 Blood flow velocity profiles at diastole and systole for proposed scenarios. Note that the warping factors and color scales differ between the shown time instants. **a** Blood flow profile at diastole ($t = T$)

for scenarios (a) *left* and (b) *right*. **b** Blood flow profile at systole ($t = 0.35 \cdot T$) for scenarios (a) *left* and (b) *right*

processes and their incorporation in real geometries continues to be matter of debate. Although residual stresses caused by prestretching have been considered in this work, the true impact of non-constitutive residual stresses is of relevance and should be addressed in the near future.

As for the model of the systemic circulation, limitations are those standard from 1D and 0D models about the calibration of arterial compliances, peripheral resistances and cardiac function.

7.3 Modeling arterial adaptations

Proper determination of internal stresses in the arterial wall is essential in the related environment in which some phenomena begin and develop. For instance, growth and remodeling processes are largely known to strongly depend on the homeostatic stress state and to be governed by the stress level in the vascular tissue (Rodríguez et al. 1994; Baek et al. 2005; Valentín and Humphrey 2009; Watton et al. 2011).

In turn, damage modeling is also acknowledged to be stress driven (Li and Robertson 2009; Alastrué et al. 2007; Balzani et al. 2012) which also is crucial in the risk of rupture assessment. Another further example is the sensitivity of mechanoreceptor function to the stretch in arterial tissue (Feng et al. 2007). Lack of arterial stretch has been proved to be related to abnormal firing rate of aortic mechanoreceptors leading to impaired ability of the central nervous system to regulate the arterial pressure. The same can be conjectured for carotid baroreceptors (featuring a similar function).

In all these situations, wrongly estimated stresses certainly would mislead the insight retrieved from models and obscure the understanding of mechanobiological arterial wall function.

From this fact is again remarked the importance of considering adequate hypotheses regarding the complete loading condition of the vessel. As seen, this can be even more critical in patient-specific cases, in which geometrical complexity

can contribute to enlarge discrepancies in the calculation of the stress state in the arterial wall.

7.4 Novel aspects

In this work, a number of modeling ingredients were considered in an integrated framework implying the consideration of the preload problem to find the material configuration on top of which constitutive equations are defined, including pressure and tethering loads, fluid–structure interaction and coupling with a dimensionally reduced model of the rest of the cardiovascular system. We remark that the integration of these modeling ingredients has no precedents; meanwhile, the issues pointed out in this work are crucial (a) to understand the interaction between the different components involved in the physical phenomena and their influence in arterial function, and (b) to accurately quantify the impact of different modeling hypothesis.

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Compliance with ethical standards

Conflicts of interest The authors declare that they have no conflict of interest.

Appendix 1: Basic relations

Consider a displacement field $\mathbf{u}_m$ which is perturbed producing $\mathbf{u}_{m,\tau} = \mathbf{u}_m + \tau \delta \mathbf{u}_m$. With this perturbation we have the deformation gradient, originally given by $\mathbf{F}_m = \mathbf{I} + \nabla_m \mathbf{u}_m$, results in $\mathbf{F}_{m,\tau} = \mathbf{I} + \nabla_m \mathbf{u}_{m,\tau} = \mathbf{I} + \nabla_m (\mathbf{u}_m + \tau \delta \mathbf{u}_m)$. Let us compute the expressions of the derivatives of several quantities involving $\mathbf{F}_{m,\tau}$, with respect to τ . This will be employed in the linearization procedures whenever the material configuration is known. Then, we have

$$\left. \frac{d}{d\tau} \mathbf{F}_{m,\tau} \right|_{\tau=0} = \nabla_m \delta \mathbf{u}_m = (\nabla_s \delta \mathbf{u}_s)_m \mathbf{F}_m, \quad (45)$$

$$\begin{aligned} \left. \frac{d}{d\tau} \mathbf{F}_{m,\tau}^{-1} \right|_{\tau=0} &= -\mathbf{F}_m^{-1} (\nabla_m \delta \mathbf{u}_m) \mathbf{F}_m^{-1} \\ &= -\mathbf{F}_m^{-1} (\nabla_s \delta \mathbf{u}_s)_m, \end{aligned} \quad (46)$$

$$\begin{aligned} \left. \frac{d}{d\tau} \det \mathbf{F}_{m,\tau} \right|_{\tau=0} &= \det \mathbf{F}_m (\mathbf{F}_m^{-T} \cdot \nabla_m \delta \mathbf{u}_m) \\ &= \det \mathbf{F}_m (\operatorname{div}_s \delta \mathbf{u}_s)_m, \end{aligned} \quad (47)$$

$$\begin{aligned} \left. \frac{d}{d\tau} \mathbf{E}_{m,\tau} \right|_{\tau=0} &= (\mathbf{F}_m^T (\nabla_m \delta \mathbf{u}_m))^S \\ &= \mathbf{F}_m^T (\nabla_s \delta \mathbf{u}_s)_m^S \mathbf{F}_m. \end{aligned} \quad (48)$$

$$\begin{aligned} \left. \frac{d}{d\tau} \left| \mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0 \right| \right|_{\tau=0} &= \left. \frac{d}{d\tau} \sqrt{\mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0 \cdot \mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0} \right|_{\tau=0} \\ &= \frac{1}{\left| \mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0 \right|} \left(-\mathbf{F}_m^{-T} (\nabla_m \delta \mathbf{u}_m)^T \mathbf{F}_m^{-T} \right) \mathbf{n}_0 \cdot \mathbf{F}_m^{-T} \mathbf{n}_0 \\ &= - \left[(\nabla_s \delta \mathbf{u}_s)^T \mathbf{n}_s \cdot \mathbf{n}_s \right]_m \left| \mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0 \right| \end{aligned} \quad (49)$$

$$\begin{aligned} \left. \frac{d}{d\tau} \left| \mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0 \right|^{-1} \right|_{\tau=0} &= - \left| \mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0 \right|^{-3} \left(-\mathbf{F}_m^{-T} (\nabla_m \delta \mathbf{u}_m)^T \mathbf{F}_m^{-T} \right) \mathbf{n}_0 \cdot \mathbf{F}_m^{-T} \mathbf{n}_0 \\ &= \left| \mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0 \right|^{-1} (\nabla_s \delta \mathbf{u}_s)_m^T (\mathbf{n}_s)_m \cdot (\mathbf{n}_s)_m \end{aligned} \quad (50)$$

$$\begin{aligned} \left. \frac{d}{d\tau} \frac{\mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0}{\left| \mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0 \right|} \right|_{\tau=0} &= -(\nabla_s \delta \mathbf{u}_s)_m^T \frac{\mathbf{F}_m^{-T} \mathbf{n}_0}{\left| \mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0 \right|} \\ &\quad + \frac{\mathbf{F}_m^{-T} \mathbf{n}_0}{\left| \mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0 \right|^2} \frac{(\nabla_s \delta \mathbf{u}_s)_m \mathbf{F}_m^{-T} \mathbf{n}_0 \cdot \mathbf{F}_m^{-T} \mathbf{n}_0}{\left| \mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0 \right|} \\ &= - \left[(\nabla_s \delta \mathbf{u}_s)^T \mathbf{n}_s \right]_m + \left[\left[(\nabla_s \delta \mathbf{u}_s)^T \mathbf{n}_s \cdot \mathbf{n}_s \right] \mathbf{n}_s \right]_m \\ &= + \left\{ \left[-(\nabla_s \delta \mathbf{u}_s)^T + \left((\nabla_s \delta \mathbf{u}_s)^T \mathbf{n}_s \cdot \mathbf{n}_s \right) \mathbf{I} \right] \mathbf{n}_s \right\}_m \end{aligned} \quad (51)$$

Now, consider the displacement field $\mathbf{u}_s$ which is perturbed producing $\mathbf{u}_{s,\tau} = \mathbf{u}_s + \tau \delta \mathbf{u}_s$. Then $\mathbf{F}_s^{-1} = \mathbf{I} - \nabla_s \mathbf{u}_s$ results in $\mathbf{F}_{s,\tau}^{-1} = \mathbf{I} - \nabla_s \mathbf{u}_{s,\tau} = \mathbf{I} - \nabla_s (\mathbf{u}_s + \tau \delta \mathbf{u}_s)$. The derivatives of several quantities involving $\mathbf{F}_{s,\tau}$ with respect to τ are

$$\left. \frac{d}{d\tau} \mathbf{F}_{s,\tau}^{-1} \right|_{\tau=0} = -\nabla_s \delta \mathbf{u}_s, \quad (52)$$

$$\left. \frac{d}{d\tau} \mathbf{F}_{s,\tau} \right|_{\tau=0} = \mathbf{F}_s (\nabla_s \delta \mathbf{u}_s) \mathbf{F}_s, \quad (53)$$

$$\left. \frac{d}{d\tau} \det \mathbf{F}_{s,\tau} \right|_{\tau=0} = \det \mathbf{F}_s (\mathbf{F}_s^T \cdot \nabla_s \delta \mathbf{u}_s), \quad (54)$$

$$\left. \frac{d}{d\tau} \mathbf{E}_{s,\tau} \right|_{\tau=0} = \mathbf{F}_s^T (\mathbf{F}_s (\nabla_s \delta \mathbf{u}_s))^S \mathbf{F}_s. \quad (55)$$

Appendix 2: Linearization procedures

Solid preload problem

For the preload problem is presented the linearization of the variational expressions (7)–(8). We recall that for this case Ω_s is fixed and for each Newton–Raphson iteration a new material configuration Ω_m^k is obtained, with points $\mathbf{x}_m^k = \mathbf{x}_s - \mathbf{u}_s^k$. In compact form the Problem 1 reads: find $(\mathbf{u}_s, \lambda_s) \in \mathcal{V}_s \times \mathcal{L}_s$ such that

$$\begin{cases} \langle \mathcal{R}_s(\mathbf{u}_s, \lambda_s), \hat{\mathbf{u}}_s \rangle_{\Omega_s} = 0 & \forall \hat{\mathbf{u}}_s \in \mathcal{V}_s \\ \langle \mathcal{J}_s(\mathbf{u}_s), \hat{\lambda}_s \rangle_{\Omega_s} = 0 & \forall \hat{\lambda}_s \in \mathcal{L}_s \end{cases} \quad (56)$$

The Newton–Raphson linearization applied to the above expression at the point $(\mathbf{u}_s^k, \lambda_s^k) \in \mathcal{U}_s \times \mathcal{L}_s$ and the increment/perturbation $(\delta \mathbf{u}_s, \delta \lambda_s) \in \mathcal{V}_s \times \mathcal{L}_s$ gives:

$$\begin{aligned} \langle \mathcal{R}_s(\mathbf{u}_s^k, \lambda_s^k), \hat{\mathbf{u}}_s \rangle_{\Omega_s} \\ + \frac{d}{d\tau} \langle \mathcal{R}_s(\mathbf{u}_s^k + \tau \delta \mathbf{u}_s, \lambda_s^k + \tau \delta \lambda_s), \hat{\mathbf{u}}_s \rangle_{\Omega_s} \Big|_{\tau=0} = 0 \quad \forall \hat{\mathbf{u}}_s \in \mathcal{V}_s \end{aligned} \quad (57)$$

$$\begin{aligned} \langle \mathcal{J}_s(\mathbf{u}_s^k), \hat{\lambda}_s \rangle_{\Omega_s} \\ + \frac{d}{d\tau} \langle \mathcal{J}_s(\mathbf{u}_s^k + \tau \delta \mathbf{u}_s), \hat{\lambda}_s \rangle_{\Omega_s} \Big|_{\tau=0} = 0 \quad \forall \hat{\lambda}_s \in \mathcal{L}_s \end{aligned} \quad (58)$$

As shown in Appendix 1, to denote the presence of the perturbation $(\tau \delta \mathbf{u}_s)$ into the quantities that depend on $\mathbf{u}_s$, we introduce the additional index τ , i.e., $\mathbf{F}_{s,\tau}^{-1} = \mathbf{I} - \nabla_s \mathbf{u}_s$ results in $\mathbf{F}_{s,\tau}^{-1} = \mathbf{I} - \nabla_s \mathbf{u}_{s,\tau} = \mathbf{I} - \nabla_s (\mathbf{u}_s + \tau \delta \mathbf{u}_s)$. For the sake of clarity, we omit index k on the fields $\mathbf{u}_s$ and λ_s (and also on quantities that are updated at each iteration such as $\mathbf{F}_s$) on the further developments. Analyzing the expanded expression for the perturbed residual in the material configuration, contribution of the second term of Eq. (57), takes the form

$$\begin{aligned} \langle \mathcal{R}_s(\mathbf{u}_s + \tau \delta \mathbf{u}_s, \lambda_s + \tau \delta \lambda_s), \hat{\mathbf{u}}_s \rangle_{\Omega_s} \\ = \int_{\Omega_s} [- (\lambda_s + \tau \delta \lambda_s) \operatorname{div} \hat{\mathbf{u}}_s + \boldsymbol{\sigma}_{s,\tau} \cdot \boldsymbol{\varepsilon}(\hat{\mathbf{u}}_s)] d\Omega_s \\ - \int_{\partial \Omega_s^W} \mathbf{P}_s \mathbf{t}_s^{W,t} \cdot \hat{\mathbf{u}}_s d\partial \Omega_s^W - \int_{\partial \Omega_s^W} t_s^{W,n} \mathbf{n}_s \cdot \hat{\mathbf{u}}_s d\partial \Omega_s^W \\ - \int_{\partial \Omega_s^N} \mathbf{t}_s^N \cdot \hat{\mathbf{u}}_s d\partial \Omega_s^N \quad \forall \hat{\mathbf{u}}_s \in \mathcal{V}_s, \end{aligned} \quad (59)$$

where the perturbed expression for the Cauchy stress tensor $\boldsymbol{\sigma}_{s,\tau}$ reads

$$\boldsymbol{\sigma}_{s,\tau} = \left(\frac{1}{\det \mathbf{F}_{s,\tau}} \mathbf{F}_{s,\tau} \mathbf{S}_{s,\tau} \mathbf{F}_{s,\tau}^T \right). \quad (60)$$

Analogously, for the second term of (58), the perturbed residuals is written as

$$\langle \mathcal{J}_s(\mathbf{u}_s + \tau \delta \mathbf{u}_s), \hat{\lambda}_s \rangle_{\Omega_s} = \int_{\Omega_s} [1 - \det \mathbf{F}_{s,\tau}^{-1}] \hat{\lambda}_s d\Omega_s = 0 \quad \forall \hat{\lambda}_s \in \mathcal{L}_s. \quad (61)$$

Using Eqs. (52)–(55) from “Appendix 1”, we obtain the derivatives of the perturbed equations

$$\begin{aligned} \frac{d}{d\tau} \langle \mathcal{R}_s(\mathbf{u}_s + \tau \delta \mathbf{u}_s, \lambda_s + \tau \delta \lambda_s), \hat{\mathbf{u}}_s \rangle_{\Omega_s} \Big|_{\tau=0} \\ = - \int_{\Omega_s} (\mathbf{F}_s^T \cdot \nabla_s \delta \mathbf{u}_s) \boldsymbol{\sigma}_s \cdot \boldsymbol{\varepsilon}(\hat{\mathbf{u}}_s) d\Omega_s \end{aligned}$$

$$\begin{aligned} + \int_{\Omega_s} 2(\mathbf{F}_s(\nabla_s \delta \mathbf{u}_s) \boldsymbol{\sigma}_s) \cdot \boldsymbol{\varepsilon}(\hat{\mathbf{u}}_s) d\Omega_s \\ + \int_{\Omega_s} \mathbf{D}_s(\mathbf{F}_s \nabla_s \delta \mathbf{u}_s)^S \cdot \boldsymbol{\varepsilon}(\hat{\mathbf{u}}_s) d\Omega_s \\ - \int_{\Omega_s} \delta \lambda_s \operatorname{div} \hat{\mathbf{u}}_s d\Omega_s, \end{aligned} \quad (62)$$

and

$$\frac{d}{d\tau} \langle \mathcal{J}_s(\mathbf{u}_s + \tau \delta \mathbf{u}_s), \hat{\lambda}_s \rangle_{\Omega_s} \Big|_{\tau=0} = - \int_{\Omega_s} (\mathbf{F}_s^T \cdot \nabla_s \delta \mathbf{u}_s) \hat{\lambda}_s d\Omega_s \quad (63)$$

where S denotes the symmetric part, and $\mathbf{D}_s(\mathbf{F}_s \nabla_s \delta \mathbf{u}_s)^S$ stands for

$$\mathbf{D}_s(\mathbf{F}_s \nabla_s \delta \mathbf{u}_s)^S = \frac{1}{\det \mathbf{F}_s} \mathbf{F}_s \left[\left(\frac{\partial \mathbf{S}_m}{\partial \mathbf{E}_m} \right)_s \mathbf{F}_s^T (\mathbf{F}_s \nabla_s \delta \mathbf{u}_s)^S \mathbf{F}_s \right] \mathbf{F}_s^T. \quad (64)$$

Then, $\mathbf{D}_s$ is a fourth-order tensor, whose components in a Cartesian system are given by

$$[\mathbf{D}_s]_{ijkl} = \frac{1}{\det \mathbf{F}_s} [\mathbf{F}_s]_{ia} [\mathbf{F}_s]_{jb} [\mathbf{F}_s]_{kc} [\mathbf{F}_s]_{ld} \left[\left(\frac{\partial \mathbf{S}_m}{\partial \mathbf{E}_m} \right)_s \right]_{abcd}. \quad (65)$$

Collecting Eqs. (62) and (63) we can now formulate the linearized problem: given $(\mathbf{u}_s, \lambda_s)$ (displacement and pressure fields on the previous iteration—omitted k index) find $(\delta \mathbf{u}_s, \delta \lambda_s) \in \mathcal{V}_s \times \mathcal{L}_s$ such that

$$\begin{cases} d_s(\delta \mathbf{u}_s, \hat{\mathbf{u}}_s) + e_s(\delta \lambda_s, \hat{\mathbf{u}}_s) = g_s(\hat{\mathbf{u}}_s) & \forall \hat{\mathbf{u}}_s \in \mathcal{V}_s \\ f_s(\delta \mathbf{u}_s, \hat{\lambda}_s) = h_s(\hat{\lambda}_s) & \forall \hat{\lambda}_s \in \mathcal{L}_s \end{cases} \quad (66)$$

with

$$\begin{aligned} d_s(\delta \mathbf{u}_s, \hat{\mathbf{u}}_s) = - \int_{\Omega_s} (\mathbf{F}_s^T \cdot \nabla_s \delta \mathbf{u}_s) \boldsymbol{\sigma}_s \cdot \boldsymbol{\varepsilon}(\hat{\mathbf{u}}_s) d\Omega_s \\ + \int_{\Omega_s} 2(\mathbf{F}_s(\nabla_s \delta \mathbf{u}_s) \boldsymbol{\sigma}_s) \cdot \boldsymbol{\varepsilon}(\hat{\mathbf{u}}_s) d\Omega_s \\ + \int_{\Omega_s} \mathbf{D}_s(\mathbf{F}_s \nabla_s \delta \mathbf{u}_s)^S \cdot \boldsymbol{\varepsilon}(\hat{\mathbf{u}}_s) d\Omega_s, \end{aligned} \quad (67)$$

$$e_s(\delta \lambda_s, \hat{\mathbf{u}}_s) = - \int_{\Omega_s} \delta \lambda_s \operatorname{div} \hat{\mathbf{u}}_s d\Omega_s, \quad (68)$$

$$\begin{aligned}
g_s(\hat{\mathbf{u}}_s) = & - \int_{\Omega_s} [-\lambda_s \operatorname{div} \hat{\mathbf{u}}_s + \boldsymbol{\sigma}_s \cdot \boldsymbol{\varepsilon}_s(\hat{\mathbf{u}}_s)] d\Omega_s \\
& + \int_{\partial\Omega_s^W} \mathbf{P}_s \mathbf{t}_s^W \cdot \hat{\mathbf{u}}_s d\partial\Omega_s^W + \int_{\partial\Omega_s^W} t_s^{W,n} \mathbf{n}_s \cdot \hat{\mathbf{u}}_s d\partial\Omega_s^W \\
& + \int_{\partial\Omega_s^N} \mathbf{t}_s^N \cdot \hat{\mathbf{u}}_s d\partial\Omega_s^N \quad (69)
\end{aligned}$$

$$f_s(\delta \mathbf{u}_s, \hat{\lambda}_s) = - \int_{\Omega_s} (\mathbf{F}_s^T \cdot \nabla_s \delta \mathbf{u}_s) \hat{\lambda}_s d\Omega_s \quad (70)$$

$$h_s(\hat{\lambda}_s) = - \int_{\Omega_s} [1 - \det \mathbf{F}_s^{-1}] \hat{\lambda}_s d\Omega_s \quad (71)$$

Solid direct problem

In this case the linearization of the variational expressions (10)–(11) is carried out taking into account that the material configuration Ω_m is fixed. In abstract form the above expressions reads: find $(\mathbf{u}_m, \lambda_m) \in \mathcal{U}_m \times \mathcal{L}_m$ such that

$$\begin{cases} \langle \mathcal{R}_m(\mathbf{u}_m, \lambda_m), \hat{\mathbf{u}}_m \rangle_{\Omega_m} = 0 & \forall \hat{\mathbf{u}}_m \in \mathcal{V}_m \\ \langle \mathcal{J}_m(\mathbf{u}_m), \hat{\lambda}_m \rangle_{\Omega_m} = 0 & \forall \hat{\lambda}_m \in \mathcal{L}_m \end{cases} \quad (72)$$

The Newton–Raphson linearization applied to the above expression at the point $(\mathbf{u}_m^k, \lambda_m^k) \in \mathcal{U}_m \times \mathcal{L}_m$ and the increment/perturbation $(\delta \mathbf{u}_m, \delta \lambda_m) \in \mathcal{V}_m \times \mathcal{L}_m$ gives:

$$\begin{aligned}
& \langle \mathcal{R}_m(\mathbf{u}_m^k, \lambda_m^k), \hat{\mathbf{u}}_m \rangle_{\Omega_m} \\
& + \frac{d}{d\tau} \langle \mathcal{R}_m(\mathbf{u}_m^k + \tau \delta \mathbf{u}_m, \lambda_m^k), \hat{\mathbf{u}}_m \rangle_{\Omega_m} \Big|_{\tau=0} \\
& + \frac{d}{d\tau} \langle \mathcal{R}_m(\mathbf{u}_m^k, \lambda_m^k + \tau \delta \lambda), \hat{\mathbf{u}}_m \rangle_{\Omega_m} \Big|_{\tau=0} \\
& = 0 \quad \forall \hat{\mathbf{u}}_m \in \mathcal{V}_m \quad (73)
\end{aligned}$$

$$\begin{aligned}
& \langle \mathcal{J}_m(\mathbf{u}_m^k), \hat{\lambda}_m \rangle_{\Omega_m} \\
& + \frac{d}{d\tau} \langle \mathcal{J}_m(\mathbf{u}_m^k + \tau \delta \mathbf{u}_m), \hat{\lambda}_m \rangle_{\Omega_m} \Big|_{\tau=0} = 0 \quad \forall \hat{\lambda}_m \in \mathcal{L}_m \quad (74)
\end{aligned}$$

As shown in “Appendix 1”, to denote the presence of the perturbation $(\tau \delta \mathbf{u}_m)$ into the quantities that depend on $\mathbf{u}$, we introduce the additional index τ , i.e., $\mathbf{F}_{m,\tau} = \mathbf{I} + \nabla_m(\mathbf{u}_m + \tau \delta \mathbf{u}_m) = \mathbf{F}_m + \tau \nabla_m \delta \mathbf{u}_m$. As before, we omit index k indicating quantities evaluated at the previous iteration. Analyzing the expanded expression for the perturbed residual in the material configuration, contribution of the second term of Eq. (73), takes the form

$$\begin{aligned}
& \frac{d}{d\tau} \langle \mathcal{R}_m(\mathbf{u}_m + \tau \delta \mathbf{u}_m, \lambda_m), \hat{\mathbf{u}}_m \rangle_{\Omega_m} \Big|_{\tau=0} \\
& = - \int_{\Omega_m} \frac{d}{d\tau} \left[\lambda_m (\mathbf{F}_{m,\tau}^{-T} \cdot \nabla_m \hat{\mathbf{u}}_m) \det \mathbf{F}_{m,\tau} \right] \Big|_{\tau=0} d\Omega_m
\end{aligned}$$

$$\begin{aligned}
& + \int_{\Omega_m} \frac{d}{d\tau} [\mathbf{S}_m(\mathbf{E}_{m,\tau}) \cdot \dot{\mathbf{E}}_\tau(\hat{\mathbf{u}}_m)] \Big|_{\tau=0} d\Omega_m \\
& - \int_{\partial\Omega_m^W} \frac{d}{d\tau} (t_m^{W,n} \mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0 \cdot \hat{\mathbf{u}}_m) \det \mathbf{F}_{m,\tau} \Big|_{\tau=0} d\partial\Omega_m^W \\
& - \int_{\partial\Omega_m^W} \frac{d}{d\tau} \mathbf{P}_{m,\tau} \mathbf{t}_m^{W,t} \cdot \hat{\mathbf{u}}_m \Big|_{\tau=0} \mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0 \Big| \det \mathbf{F}_{m,\tau} \Big|_{\tau=0} d\partial\Omega_m^W \\
& - \int_{\partial\Omega_m^N} \frac{d}{d\tau} (\mathbf{t}_m^N \cdot \hat{\mathbf{u}}_m) \Big|_{\tau=0} \mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0 \Big| \det \mathbf{F}_{m,\tau} \Big|_{\tau=0} d\partial\Omega_m^N, \quad (75)
\end{aligned}$$

where $\dot{\mathbf{E}}_\tau(\hat{\mathbf{u}}_m) = \frac{1}{2} [\mathbf{F}_{m,\tau}^T (\nabla_m \hat{\mathbf{u}}_m) + (\nabla_m \hat{\mathbf{u}}_m)^T \mathbf{F}_{m,\tau}]$. Using Eqs. (45)–(51) detailed in “Appendix 1” to perform the derivation with respect to τ (and evaluate at $\tau = 0$) the following expression is obtained

$$\begin{aligned}
& \frac{d}{d\tau} \langle \mathcal{R}_m(\mathbf{u}_m + \tau \delta \mathbf{u}_m, \lambda_m), \hat{\mathbf{u}}_m \rangle_{\Omega_m} \Big|_{\tau=0} \\
& = \int_{\Omega_m} \lambda_m \left[(\mathbf{F}_m^{-T} (\nabla_m \delta \mathbf{u}_m)^T \mathbf{F}_m^{-T}) \cdot (\nabla_m \hat{\mathbf{u}}_m) \right] \det \mathbf{F}_m d\Omega_m \\
& - \int_{\Omega_m} \lambda_m \left[(\mathbf{F}_m^{-T} \cdot (\nabla_m \delta \mathbf{u}_m)) (\mathbf{F}_m^{-T} \cdot (\nabla_m \hat{\mathbf{u}}_m)) \right] \det \mathbf{F}_m d\Omega_m \\
& + \int_{\Omega_m} \left(\frac{\partial \mathbf{S}_m}{\partial \mathbf{E}_m} ((\nabla_m \delta \mathbf{u}_m)^T \mathbf{F}_m)^S \right) \cdot \dot{\mathbf{E}}(\hat{\mathbf{u}}_m) d\Omega_m \\
& + \int_{\Omega_m} \mathbf{S}_m(\mathbf{E}_m) \cdot ((\nabla_m \delta \mathbf{u}_m)^T (\nabla_m \hat{\mathbf{u}}_m))^S d\Omega_m \\
& - \int_{\partial\Omega_m^W} (t_m^{W,n} \mathbf{F}_m^{-T} \mathbf{n}_0 \cdot \hat{\mathbf{u}}_m) \det \mathbf{F}_m (\mathbf{F}_m^{-T} \cdot \nabla_m \delta \mathbf{u}_m) d\partial\Omega_m^W \\
& - \int_{\partial\Omega_m^W} \left[t_m^{W,n} \frac{d}{d\tau} (\mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0) \Big|_{\tau=0} \cdot \hat{\mathbf{u}}_m \right] \det \mathbf{F}_m d\partial\Omega_m^W \\
& - \int_{\partial\Omega_m^W} \frac{d}{d\tau} \mathbf{P}_{m,\tau} \Big|_{\tau=0} \mathbf{t}_m^{W,t} \cdot \hat{\mathbf{u}}_m \Big|_{\tau=0} \mathbf{F}_m^{-T} \mathbf{n}_0 \Big| \det \mathbf{F}_m d\partial\Omega_m^W \\
& - \int_{\partial\Omega_m^W} \mathbf{P}_m \mathbf{t}_m^{W,t} \cdot \hat{\mathbf{u}}_m \Big|_{\tau=0} \mathbf{F}_m^{-T} \mathbf{n}_0 \Big| \det \mathbf{F}_m (\mathbf{F}_m^{-T} \cdot \nabla_m \delta \mathbf{u}_m) d\partial\Omega_m^W \\
& - \int_{\partial\Omega_m^W} \mathbf{P}_m \mathbf{t}_m^{W,t} \cdot \hat{\mathbf{u}}_m \frac{\frac{d}{d\tau} \mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0 \Big|_{\tau=0} \cdot \mathbf{F}_m^{-T} \mathbf{n}_0}{\left| \mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0 \right|} \det \mathbf{F}_m d\partial\Omega_m^W \\
& - \int_{\partial\Omega_m^N} (\mathbf{t}_m^N \cdot \hat{\mathbf{u}}_m) \Big|_{\tau=0} \mathbf{F}_m^{-T} \mathbf{n}_0 \Big| \det \mathbf{F}_m (\mathbf{F}_m^{-T} \cdot \nabla_m \delta \mathbf{u}_m) d\partial\Omega_m^N \\
& - \int_{\partial\Omega_m^N} (\mathbf{t}_m^N \cdot \hat{\mathbf{u}}_m) \frac{\frac{d}{d\tau} (\mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0) \Big|_{\tau=0} \cdot \mathbf{F}_m^{-T} \mathbf{n}_0}{\left| \mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0 \right|} \det \mathbf{F}_m d\partial\Omega_m^N \quad (76)
\end{aligned}$$

where $\frac{d}{d\tau} \mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0|_{\tau=0} = (-\mathbf{F}_m^{-T} (\nabla_m \delta \mathbf{u}_m)^T \mathbf{F}_m^{-T}) \mathbf{n}_0$ and

$$\frac{d}{d\tau} \mathbf{P}_{m,\tau}|_{\tau=0} = \left[\frac{d}{d\tau} \frac{\mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0}{|\mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0|} \bigg|_{\tau=0} \otimes \frac{\mathbf{F}_m^{-T} \mathbf{n}_0}{|\mathbf{F}_m^{-T} \mathbf{n}_0|} \right]^S,$$

with $\frac{\mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0}{|\mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0|} \bigg|_{\tau=0}$ given by expression (51) provided in “Appendix 1”. Similarly, for the contribution of the third term of (73), a perturbation $(\tau \delta \lambda_m)$ is introduced, and the expression is derived. The calculation of this block is straightforward since $\langle \mathcal{R}_m, \hat{\mathbf{u}}_m \rangle_{\Omega_m}$ is linear in λ_m , then

$$\begin{aligned} \frac{d}{d\tau} \langle \mathcal{R}_m(\mathbf{u}_m, \lambda_m + \tau \delta \lambda_m), \hat{\mathbf{u}}_m \rangle_{\Omega_m} \bigg|_{\tau=0} \\ = - \int_{\Omega_m} \delta \lambda_m \left(\mathbf{F}_m^{-T} \cdot \nabla_m \hat{\mathbf{u}}_m \right) \det \mathbf{F}_m d\Omega_m. \end{aligned} \quad (77)$$

Finally, the second term of (74) evaluated at $\tau = 0$ yields

$$\begin{aligned} \frac{d}{d\tau} \langle \mathcal{J}_m(\mathbf{u}_m + \tau \delta \mathbf{u}_m), \hat{\lambda}_m \rangle_{\Omega_m} \bigg|_{\tau=0} \\ = \int_{\Omega_m} \det \mathbf{F}_m (\mathbf{F}_m^{-T} \cdot \nabla_m \delta \mathbf{u}_m) \hat{\lambda}_m d\Omega_m. \end{aligned} \quad (78)$$

With these blocks we are able to write the linear problem: given $(\mathbf{u}_m, \lambda_m)$ (displacement and pressure fields at previous iteration—omitted k index), find $(\delta \mathbf{u}_m, \delta \lambda_m) \in \mathcal{V}_m \times \mathcal{L}_m$ such that

$$\begin{cases} a_m(\delta \mathbf{u}_m, \hat{\mathbf{u}}_m) + b_m(\delta \lambda_m, \hat{\mathbf{u}}_m) = l_m(\hat{\mathbf{u}}_m) & \forall \hat{\mathbf{u}}_m \in \mathcal{V}_m \\ c_m(\delta \mathbf{u}_m, \hat{\lambda}_m) = m_m(\hat{\lambda}_m) & \forall \hat{\lambda}_m \in \mathcal{L}_m \end{cases} \quad (79)$$

where the linear and bilinear forms are obtained through

$$\begin{aligned} l_m(\hat{\mathbf{u}}_m) &= \int_{\Omega_m} \lambda_m \left(\mathbf{F}_m^{-T} \cdot \nabla_m \hat{\mathbf{u}}_m \right) \det \mathbf{F}_m d\Omega_m \\ &\quad - \int_{\Omega_m} \left(\boldsymbol{\Sigma}_m^R + \mathbf{S}_m(\mathbf{E}_m) \right) \cdot \dot{\mathbf{E}}(\hat{\mathbf{u}}_m) d\Omega_m \\ &\quad + \int_{\partial \Omega_m^W} \left(\mathbf{t}_m^{W,n} \mathbf{F}_m^{-T} \mathbf{n}_0 \cdot \hat{\mathbf{u}}_m \right) \det \mathbf{F}_m d\partial \Omega_m^W \\ &\quad + \int_{\partial \Omega_m^W} \mathbf{P}_m \mathbf{t}_m^{W,t} \cdot \hat{\mathbf{u}}_m \frac{\det \mathbf{F}_m}{|\mathbf{F}_m^{-T} \mathbf{n}_0|} d\partial \Omega_m^W \\ &\quad + \int_{\partial \Omega_m^N} \left(\mathbf{t}_m^N \cdot \hat{\mathbf{u}}_m \right) |\mathbf{F}_m^{-T} \mathbf{n}_0| \det \mathbf{F}_m d\partial \Omega_m^N, \end{aligned} \quad (80)$$

$$\begin{aligned} a_m((\delta \mathbf{u}_m, \delta \lambda_m), \hat{\mathbf{u}}_m) \\ = \int_{\Omega_m} \lambda_m \left[\left(\mathbf{F}_m^{-T} (\nabla_m \delta \mathbf{u}_m)^T \mathbf{F}_m^{-T} \right) \cdot (\nabla_m \hat{\mathbf{u}}_m) \right] \det \mathbf{F}_m d\Omega_m \\ - \int_{\Omega_m} \lambda_m \left[\left(\mathbf{F}_m^{-T} \cdot (\nabla_m \delta \mathbf{u}_m) \right) \left(\mathbf{F}_m^{-T} \cdot (\nabla_m \hat{\mathbf{u}}_m) \right) \right] \det \mathbf{F}_m d\Omega_m \\ + \int_{\Omega_m} \left(\frac{\partial \mathbf{S}_m}{\partial \mathbf{E}_m} \left((\nabla_m \delta \mathbf{u}_m)^T \mathbf{F}_m \right)^S \right) \cdot \dot{\mathbf{E}}(\hat{\mathbf{u}}_m) d\Omega_m \\ + \int_{\Omega_m} \mathbf{S}_m(\mathbf{E}_m) \cdot \left((\nabla_m \delta \mathbf{u}_m)^T (\nabla_m \hat{\mathbf{u}}_m) \right)^S d\Omega_m \\ - \int_{\partial \Omega_m^W} \left(\mathbf{t}_m^{W,n} \mathbf{F}_m^{-T} \mathbf{n}_0 \cdot \hat{\mathbf{u}}_m \right) \det \mathbf{F}_m \left(\mathbf{F}_m^{-T} \cdot \nabla_m \delta \mathbf{u}_m \right) d\partial \Omega_m^W \\ - \int_{\partial \Omega_m^W} \left[\mathbf{t}_m^{W,n} \frac{d}{d\tau} \left(\mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0 \right) \bigg|_{\tau=0} \cdot \hat{\mathbf{u}}_m \right] \det \mathbf{F}_m d\partial \Omega_m^W \\ - \int_{\partial \Omega_m^W} \frac{d}{d\tau} \mathbf{P}_{m,\tau}|_{\tau=0} \mathbf{t}_m^{W,t} \cdot \hat{\mathbf{u}}_m |\mathbf{F}_m^{-T} \mathbf{n}_0| \det \mathbf{F}_m d\partial \Omega_m^W \\ - \int_{\partial \Omega_m^W} \mathbf{P}_m \mathbf{t}_m^{W,t} \cdot \hat{\mathbf{u}}_m |\mathbf{F}_m^{-T} \mathbf{n}_0| \det \mathbf{F}_m \left(\mathbf{F}_m^{-T} \cdot \nabla_m \delta \mathbf{u}_m \right) d\partial \Omega_m^W \\ - \int_{\partial \Omega_m^W} \mathbf{P}_m \mathbf{t}_m^{W,t} \cdot \hat{\mathbf{u}}_m \frac{\frac{d}{d\tau} \mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0|_{\tau=0} \cdot \mathbf{F}_m^{-T} \mathbf{n}_0}{|\mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0|} \det \mathbf{F}_m d\partial \Omega_m^W \\ - \int_{\partial \Omega_m^N} \left(\mathbf{t}_m^N \cdot \hat{\mathbf{u}}_m \right) |\mathbf{F}_m^{-T} \mathbf{n}_0| \det \mathbf{F}_m \left(\mathbf{F}_m^{-T} \cdot \nabla_m \delta \mathbf{u}_m \right) d\partial \Omega_m^N \\ - \int_{\partial \Omega_m^N} \left(\mathbf{t}_m^N \cdot \hat{\mathbf{u}}_m \right) \frac{\frac{d}{d\tau} \left(\mathbf{F}_{m,\tau}^{-T} \mathbf{n}_0 \right) \bigg|_{\tau=0} \cdot \mathbf{F}_m^{-T} \mathbf{n}_0}{|\mathbf{F}_m^{-T} \mathbf{n}_0|} \det \mathbf{F}_m d\partial \Omega_m^N, \end{aligned} \quad (81)$$

$$b_m(\delta \lambda_m, \hat{\mathbf{u}}_m) = - \int_{\Omega_m} \delta \lambda_m \left(\mathbf{F}_m^{-T} \cdot \nabla_m \hat{\mathbf{u}}_m \right) \det \mathbf{F}_m d\Omega_m, \quad (82)$$

$$c_m(\delta \mathbf{u}_m, \hat{\lambda}_m) = \int_{\Omega_m} \det \mathbf{F}_m \left(\mathbf{F}_m^{-T} \cdot \nabla_m \delta \mathbf{u}_m \right) \hat{\lambda}_m d\Omega_m, \quad (83)$$

and

$$m_m(\hat{\lambda}_m) = - \int_{\Omega_m} (\det \mathbf{F}_m - 1) \hat{\lambda}_m d\Omega_m \quad (84)$$

The above linear problem is convenient to be rewritten (evaluated) in terms of the variables now defined in an updated configuration Ω_s^k (written as Ω_s for sake of simplicity), with points $x_s^k = x_m + u_m^k$. To do that as a first step we seek for the spatial expression of the tangent components. With the expressions provided in “Appendix 1” in mind, we can write the spatial version of (76) as

$$\begin{aligned}
& \frac{d}{d\tau} \langle \mathcal{R}_s(\mathbf{u}_s^k + \tau \delta \mathbf{u}_s, \lambda_s), \hat{\mathbf{u}}_s \rangle_{\Omega_s} \Big|_{\tau=0} \\
&= \int_{\Omega_s} \lambda_s ((\nabla_s \delta \mathbf{u}_s)^T \cdot (\nabla_s \hat{\mathbf{u}}_s) - (\operatorname{div}_s \delta \mathbf{u}_s)(\operatorname{div}_s \hat{\mathbf{u}}_s)) \, d\Omega_s \\
&+ \int_{\Omega_s} (\mathbf{D}_s \boldsymbol{\varepsilon}(\delta \mathbf{u}_s) \cdot \boldsymbol{\varepsilon}(\hat{\mathbf{u}}_s) + (\nabla_s \delta \mathbf{u}_s) \sigma_s \cdot (\nabla_s \hat{\mathbf{u}}_s)) \, d\Omega_s, \\
&\int_{\partial \Omega_m^W} \left[t_s^{W,n} \left((\nabla_s \delta \mathbf{u}_s)^T - \mathbf{I} \right) \mathbf{n}_s \cdot \hat{\mathbf{u}}_s \right] \operatorname{div}_s \delta \mathbf{u}_s \, d\partial \Omega_s^W \\
&- \int_{\partial \Omega_s^W} (\mathbf{H}(\delta \mathbf{u}_s) \mathbf{n}_s \otimes \mathbf{n}_s)^S \mathbf{t}_s^{W,t} \cdot \hat{\mathbf{u}}_s \, d\partial \Omega_s^W \\
&- \int_{\partial \Omega_m^W} \mathbf{P}_s \mathbf{t}_s^{W,t} \cdot \hat{\mathbf{u}}_s (\operatorname{div}_s \delta \mathbf{u}_s) \, d\partial \Omega_s^W \\
&- \int_{\partial \Omega_m^W} \mathbf{P}_s \mathbf{t}_s^{W,t} \cdot \hat{\mathbf{u}}_s \left(-(\nabla_s \delta \mathbf{u}_s)^T \mathbf{n}_s \cdot \mathbf{n}_s \right) \, d\partial \Omega_s^W \\
&- \int_{\partial \Omega_s^N} \left(\mathbf{t}_s^N \cdot \hat{\mathbf{u}}_s \right) \left[(\operatorname{div}_s \delta \mathbf{u}_s) - (\nabla_s \delta \mathbf{u}_s)^T \mathbf{n}_s \cdot \mathbf{n}_s \right] \, d\partial \Omega_s^N
\end{aligned} \tag{85}$$

where $\mathbf{H}(\delta \mathbf{u}_s)$ stands for

$$\mathbf{H}(\delta \mathbf{u}_s) = \left[-(\nabla_s \delta \mathbf{u}_s)^T + \left((\nabla_s \delta \mathbf{u}_s)^T \mathbf{n}_s \cdot \mathbf{n}_s \right) \mathbf{I} \right], \tag{86}$$

and recalling (65), $\mathbf{D}_s \boldsymbol{\varepsilon}_s(\delta \mathbf{u}_s)$ is given by

$$\mathbf{D}_s \boldsymbol{\varepsilon}_s(\delta \mathbf{u}_s) = \frac{1}{\det \mathbf{F}_s} \mathbf{F}_s \left[\left(\frac{\partial \mathbf{S}_m}{\partial \mathbf{E}_m} \right)_s \mathbf{F}_s^T \boldsymbol{\varepsilon}(\delta \mathbf{u}_s) \mathbf{F}_s \right] \mathbf{F}_s^T. \tag{87}$$

Now we add and subtract the term $\lambda_s (\nabla_s \delta \mathbf{u}_s) \cdot (\nabla_s \hat{\mathbf{u}}_s)$ into Eq. (85) and consider the operation with second order tensors

$$\lambda_s (\operatorname{div} \delta \mathbf{u}_s) (\operatorname{div} \hat{\mathbf{u}}_s) = \lambda_s (\mathbf{I} \otimes \mathbf{I}) \boldsymbol{\varepsilon}(\delta \mathbf{u}_s) \cdot \boldsymbol{\varepsilon}(\hat{\mathbf{u}}_s), \tag{88}$$

leading to

$$\begin{aligned}
& \frac{d}{d\tau} \langle \mathcal{R}_s(\mathbf{u}_s + \tau \delta \mathbf{u}_s, \lambda_s), \hat{\mathbf{u}}_s \rangle_{\Omega_s} \Big|_{\tau=0} \\
&= \int_{\Omega_s} (\mathbf{D}_s + \lambda_s (2\mathbb{I} - (\mathbf{I} \otimes \mathbf{I}))) \boldsymbol{\varepsilon}(\delta \mathbf{u}_s) \cdot \boldsymbol{\varepsilon}(\hat{\mathbf{u}}_s) \, d\Omega_s \\
&+ \int_{\Omega_s} (\nabla_s \delta \mathbf{u}_s) \sigma_s \cdot (\nabla_s \hat{\mathbf{u}}_s) \, d\Omega_s \\
&+ \int_{\partial \Omega_m^W} \left[t_s^{W,n} \left((\nabla_s \delta \mathbf{u}_s)^T - \mathbf{I} \right) \mathbf{n}_s \cdot \hat{\mathbf{u}}_s \right] \operatorname{div}_s \delta \mathbf{u}_s \, d\partial \Omega_s^W
\end{aligned}$$

$$\begin{aligned}
& - \int_{\partial \Omega_s^W} (\mathbf{H}(\delta \mathbf{u}_s) \mathbf{n}_s \otimes \mathbf{n}_s)^S \mathbf{t}_s^{W,t} \cdot \hat{\mathbf{u}}_s \, d\partial \Omega_s^W \\
& - \int_{\partial \Omega_m^W} \mathbf{P}_s \mathbf{t}_s^{W,t} \cdot \hat{\mathbf{u}}_s (\operatorname{div}_s \delta \mathbf{u}_s) \, d\partial \Omega_s^W \\
& - \int_{\partial \Omega_m^W} \mathbf{P}_s \mathbf{t}_s^{W,t} \cdot \hat{\mathbf{u}}_s \left(-(\nabla_s \delta \mathbf{u}_s)^T \mathbf{n}_s \cdot \mathbf{n}_s \right) \, d\partial \Omega_s^W \\
& - \int_{\partial \Omega_s^N} \left(\mathbf{t}_s^N \cdot \hat{\mathbf{u}}_s \right) \left[(\operatorname{div}_s \delta \mathbf{u}_s) - (\nabla_s \delta \mathbf{u}_s)^T \mathbf{n}_s \cdot \mathbf{n}_s \right] \, d\partial \Omega_s^N
\end{aligned} \tag{89}$$

where $\mathbb{I}$ and $\mathbf{I}$ are the fourth and second order identity tensors, respectively. Analogously, the spatial expressions (77) and (78) result in

$$\begin{aligned}
& \frac{d}{d\tau} \langle \mathcal{R}_s(\mathbf{u}_s, \lambda_s + \tau \delta \lambda_s), \hat{\mathbf{u}}_s \rangle_{\Omega_s} \Big|_{\tau=0} \\
&= - \int_{\Omega_s} \delta \lambda_s \operatorname{div} \hat{\mathbf{u}}_s \, d\Omega_s,
\end{aligned} \tag{90}$$

and

$$\begin{aligned}
& \frac{d}{d\tau} \langle \mathcal{J}_s(\mathbf{u}_s + \tau \delta \mathbf{u}_s, \lambda_s), \hat{\lambda}_s \rangle_{\Omega_s} \Big|_{\tau=0} \\
&= \int_{\Omega_s} \operatorname{div} \delta \mathbf{u}_s \hat{\lambda}_s \, d\Omega_s.
\end{aligned} \tag{91}$$

Hence, from Eqs. (85), (90) and (91), the spatial form of the linearized problem for incompressible materials is formulated as follows: given $(\mathbf{u}_s, \lambda_s)$ (displacement and pressure fields at previous Newton–Raphson iteration-omitted k index-) find $(\delta \mathbf{u}_s, \delta \lambda_s) \in \mathcal{V}_s \times \mathcal{L}_s$ such that

$$\begin{cases} a_s(\delta \mathbf{u}_s, \hat{\mathbf{u}}_s) + b_s(\delta \lambda_s, \hat{\mathbf{u}}_s) = l_s(\hat{\mathbf{u}}_s) & \forall \hat{\mathbf{u}}_s \in \mathcal{V}_s \\ c_s(\delta \mathbf{u}_s, \hat{\lambda}_s) = m_s(\hat{\lambda}_s) & \forall \hat{\lambda}_s \in \mathcal{L}_s \end{cases} \tag{92}$$

where the bilinear and linear forms are given by

$$\begin{aligned}
& a_s(\delta \mathbf{u}_s, \hat{\mathbf{u}}_s) \\
&= \int_{\Omega_s} [\mathbf{D}_s + \lambda_s (2\mathbb{I} - (\mathbf{I} \otimes \mathbf{I}))] \boldsymbol{\varepsilon}(\delta \mathbf{u}_s) \cdot \boldsymbol{\varepsilon}(\hat{\mathbf{u}}_s) \, d\Omega_s \\
&+ \int_{\Omega_s} (\nabla_s \delta \mathbf{u}_s) \sigma_s \cdot (\nabla_s \hat{\mathbf{u}}_s) \, d\Omega_s \\
&+ \int_{\partial \Omega_m^W} \left[t_s^{W,n} ((\nabla_s \delta \mathbf{u}_s)^T - \mathbf{I}) \mathbf{n}_s^W \cdot \hat{\mathbf{u}}_s \right] \operatorname{div}_s \delta \mathbf{u}_s \, d\partial \Omega_s^W \\
&- \int_{\partial \Omega_s^W} (\mathbf{H}(\delta \mathbf{u}_s) \mathbf{n}_s \otimes \mathbf{n}_s)^S \mathbf{t}_s^{W,t} \cdot \hat{\mathbf{u}}_s \, d\partial \Omega_s^W
\end{aligned}$$

$$\begin{aligned}
& - \int_{\partial\Omega_m^W} \mathbf{P}_s \mathbf{t}_s^{W,t} \cdot \hat{\mathbf{u}}_s (\operatorname{div}_s \delta \mathbf{u}_s) \, d\partial\Omega_s^W \\
& - \int_{\partial\Omega_m^W} \mathbf{P}_s \mathbf{t}_s^{W,t} \cdot \hat{\mathbf{u}}_s \left(-(\nabla_s \delta \mathbf{u}_s)^T \mathbf{n}_s \cdot \mathbf{n}_s \right) \, d\partial\Omega_s^W \\
& - \int_{\partial\Omega_s^N} \left(\mathbf{t}_s^N \cdot \hat{\mathbf{u}}_s \right) \left[(\operatorname{div}_s \delta \mathbf{u}_s) - (\nabla_s \delta \mathbf{u}_s)^T \mathbf{n}_s \cdot \mathbf{n}_s \right] \, d\partial\Omega_s^N,
\end{aligned} \quad (93)$$

$$b_s(\delta \lambda_s, \hat{\mathbf{u}}_s) = - \int_{\Omega_s} \delta \lambda_s \operatorname{div} \hat{\mathbf{u}}_s \, d\Omega_s, \quad (94)$$

$$\begin{aligned}
l_s(\hat{\mathbf{u}}_s) &= - \int_{\Omega_s} \left[-\lambda_s \operatorname{div} \hat{\mathbf{u}}_s + \boldsymbol{\sigma}_s \cdot \boldsymbol{\varepsilon}_s(\hat{\mathbf{u}}_s) \right] \, d\Omega_s \\
&+ \int_{\partial\Omega_s^W} \mathbf{P}_s \mathbf{t}_s^W \cdot \hat{\mathbf{u}}_s \, d\partial\Omega_s^W + \int_{\partial\Omega_s^W} t_s^{W,n} \mathbf{n}_s \cdot \hat{\mathbf{u}}_s \, d\partial\Omega_s^W \\
&+ \int_{\partial\Omega_s^N} \mathbf{t}_s^N \cdot \hat{\mathbf{u}}_s \, d\partial\Omega_s^N,
\end{aligned} \quad (95)$$

$$c_s(\delta \mathbf{u}_s, \hat{\lambda}_s) = \int_{\Omega_s} \operatorname{div} \delta \hat{\mathbf{u}}_s \hat{\lambda}_s \, d\Omega_s, \quad (96)$$

and

$$m_s(\hat{\lambda}_s) = - \int_{\Omega_s} \left(1 - \det \mathbf{F}_s^{-1} \right) \hat{\lambda}_s \, d\Omega_s. \quad (97)$$

Fluid problem

In order to solve numerically the variational problem set by Eqs. (30)–(32), a fixed-point method is used where the velocity of the reference domain at each iteration is calculated as $\mathbf{v}^k = \frac{\mathbf{d}^k - \mathbf{d}^n}{\Delta t}$ (other choices are perfectly viable), with superscripts k and n referring to the solution at the previous fixed-point iteration and at the previous time step, respectively. The proposed fixed-point method reads: until convergence is achieved, for every iteration $k = 0, 1, \dots$ find $(\mathbf{v}^{k+1}, p^{k+1}, \mathbf{d}^{k+1}) \in \mathcal{V}_t^k \times \mathcal{P}_t^k \times \mathcal{D}_t^k$ such that

$$\begin{aligned}
& \int_{\gamma_t^k} \rho \frac{\partial \mathbf{v}^{k+1}}{\partial t} \cdot \hat{\mathbf{v}} \, d\gamma_t^k + \int_{\gamma_t^k} \rho \nabla \mathbf{v}^{k+1} \left(\mathbf{v}^k - \mathbf{w}^k \right) \cdot \hat{\mathbf{v}} \, d\gamma_t^k \\
& - \int_{\gamma_t^k} p^{k+1} \operatorname{div} \hat{\mathbf{v}} \, d\gamma_t^k + \int_{\gamma_t^k} \bar{\boldsymbol{\sigma}} \left(\mathbf{v}^{k+1} \right) \cdot \boldsymbol{\varepsilon}(\hat{\mathbf{v}}) \, d\gamma_t^k \\
& = \sum_{i=1}^C \int_{\partial\gamma_t^{k,A,i}} t_t^i \mathbf{n}^i \cdot \hat{\mathbf{v}} \, d\partial\gamma_t^{k,A,i} \quad \forall \hat{\mathbf{v}} \in \mathcal{V}_t^{k,*},
\end{aligned} \quad (98)$$

$$\int_{\gamma_t^k} \hat{p} \operatorname{div} \mathbf{v}^{k+1} \, d\gamma_t^k = 0 \quad \forall \hat{p} \in \mathcal{P}_t^k, \quad (99)$$

$$\int_{\gamma_t^k} \nabla \mathbf{d}^{k+1} \cdot \nabla \hat{\mathbf{d}} \, d\gamma_t^k = 0 \quad \forall \hat{\mathbf{d}} \in \mathcal{D}_t^{k,*}, \quad (100)$$

where $\mathcal{V}_t^{k,*}$, $\mathcal{P}_t^k$ and $\mathcal{D}_t^{k,*}$ are the counterparts of $\mathcal{V}_t^*$, $\mathcal{P}_t$ and $\mathcal{D}_t^*$ but with functions defined over γ_t^k . Considering a Crank–Nicolson scheme for the time integration, the time derivative is approximated as follows

$$\frac{\partial \mathbf{v}^{k+1}}{\partial t} = \frac{\mathbf{v}^{k+1} - \mathbf{v}^n}{\Delta t}, \quad (101)$$

and all other terms in (98), except for the coupling with the pressure field, are evaluated at $t_{n+\frac{1}{2}} = \frac{1}{2}(t_{n+1} + t_n)$.

References

- Alastrué V, Rodríguez J, Calvo B, Doblaré M (2007) Structural damage models for fibrous biological soft tissues. *Int J Solid Struct* 44(18–19):5894–5911
- Alastrué V, Peña E, Martínez MA, Doblaré M (2007) Assessing the use of the opening angle method to enforce residual stresses in patient-specific arteries. *Ann Biomed Eng* 35(10):1821–1837
- Avolio A (1980) Multi-branched model of the human arterial system. *Med Biol Eng Comput* 18:709–718
- Baek S, Rajagopal K, Humphrey J (2005) Competition between radial expansion and thickening in the enlargement of an intracranial saccular aneurysm. *J Elast* 80(1–3):13–31
- Balzani D, Brinkhues S, Holzapfel Ga (2012) Constitutive framework for the modeling of damage in collagenous soft tissues with application to arterial walls. *Comput Methods Appl Mech Eng* 213–216:139–151
- Bharadvaj B, Mabon R, Giddens D (1982) Steady flow in a model of the human carotid bifurcation. Part I: flow visualization. *J Biomech* 15(5):349–362
- Blanco PJ, Feijóo RA, Urquiza SA (2007) A unified variational approach for coupling 3D–1D models and its blood flow applications. *Comput Methods Appl Mech Eng* 41–44:4391–4410
- Blanco PJ, Pivello MR, Urquiza SA, Feijóo RA (2009) On the potentialities of 3D–1D coupled models in hemodynamics simulations. *J Biomech* 42(7):919–930
- Blanco PJ, Urquiza SA, Feijóo RA (2010) Assessing the influence of heart rate in local hemodynamics through coupled 3D–1D–0D models. *Int J Numer Methods Biomed Eng* 26:890–903
- Blanco PJ, Deparis S, Malossi ACI (2013a) On the continuity of mean total normal stress in geometrical multiscale cardiovascular problems. *J Comput Phys* 251:136–155
- Blanco PJ, Leiva JS, Buscaglia GC (2013b) A black-box decomposition approach for coupling heterogeneous components in hemodynamics simulations. *Int J Numer Methods Biomed Eng* 29(3):408–427
- Blanco PJ, Feijóo RA (2013) A dimensionally-heterogeneous closed-loop model for the cardiovascular system and its applications. *Med Eng Phys* 35:652–667
- Castro MA, Putman CM, Cebal JR (2011) Computational analysis of anterior communicating artery aneurysm shear stress before and after aneurysm formation. *J Phys Conf Ser* 332(1):012001

- Crosetto P, Reymond P, Deparis S, Kontarakis D, Stergiopoulos N, Quarteroni A (2011) Fluid–structure interaction simulation of aortic blood flow. *Comput. Fluids* 43(1):46–57
- Dempere-Marco L, Oubel E, Castro M, Putman C, Frangi A, Cebal J (2006) Cfd analysis incorporating the influence of wall motion: application to intracranial aneurysms. *Medical image computing and computer-assisted intervention MICCAI 2006*:438–445
- Deparis S, Fernandez M, Formaggia L, Nobile F (2003) Acceleration of a fixed point algorithm for fluid–structure interaction using transpiration conditions. In: Bathe KJ (ed) *Computational fluid and solid mechanics 2003*. Elsevier Science Ltd, Oxford, pp 1307–1311
- Deparis S, Discacciati M, Fourestey G, Quarteroni A (2006) Fluid–structure algorithms based on Steklov–Poincaré operators. *Comput Methods Appl Mech Eng* 195(41–43):5797–5812
- Donea J, Huerta A (2003) *Finite element methods for flow problems*. Wiley, Chichester
- Feng B, Li B, Nauman EA, Schild JH (2007) Theoretical and electrophysiological evidence for axial loading about aortic baroreceptor nerve terminals in rats. *Am J Phys Heart Circ Physiol* 293(6):H3659–H3672
- Formaggia L, Nobile F, Quarteroni A, Veneziani A (1999) Multiscale modelling of the circulatory system: a preliminary analysis. *Comput Vis Sci* 2(2–3):75–83
- Formaggia L, Gerbeau J, Nobile F, Quarteroni A (2001) On the coupling of 3D and 1D Navier–Stokes equations for flow problems in compliant vessels. *Comput Methods Appl Mech Eng* 191:561–582
- Formaggia L, Quarteroni A, Vergara C (2013) On the physical consistency between three-dimensional and one-dimensional models in haemodynamics. *J Comput Phys* 244:97–112
- Franca LP, Hughes TJ (1988) Two classes of mixed finite element methods. *Comput Methods Appl Mech Eng* 59:89–129
- Gasser TC, Ogden RW, Holzapfel GA (2006) Hyperelastic modelling of arterial layers with distributed collagen fibre orientations. *J R Soc Interface* 3(6):15–35
- Gee M, Reeps C, Eckstein H, Wall W (2009) Prestressing in finite deformation abdominal aortic aneurysm simulation. *J Biomech* 42:1732–1739
- Gee MW, Forster C, Wall WA (2010) A computational strategy for prestressing patient-specific biomechanical problems under finite deformation. *Int J Numer Methods Biomed Eng* 26(1):52–72
- Govindjee S, Mihalic PA (1996) Computational methods for inverse finite elastostatics. *Comput Methods Appl Mech Eng* 136(1–2):47–57
- Govindjee S, Mihalic PA (1998) Computational methods for inverse deformations in quasi-incompressible finite elasticity. *Int J Numer Methods Eng* 43(5):821–838
- Grinberg L, Karniadakis GE (2008) Outflow boundary conditions for arterial networks with multiple outlets. *Ann Biomed Eng* 36(9):1496–1514
- Heidenreich PA, Trogon JG, Khavjou OA, Butler J, Dracup K, Ezekowitz M, Finkelstein EA, Hong Y, Johnston ALJD, Cand Khera S, Lloyd-Jones D, Nelson SA, Nichol G, Orenstein D, Wilson PW, Woo YJ (2011) Forecasting the future of cardiovascular disease in the United States: a policy statement from the American heart association. *Circulation* 123(8):933–944
- Holzapfel GA, Gasser TC (2000) A new constitutive framework for arterial wall mechanics and a comparative study of material models. *J Elast* 61:1–48
- Holzapfel GA, Ogden R (2010a) Constitutive modelling of arteries. *Proc R Soc A Math Phys Eng Sci* 466(2118):1551–1597
- Holzapfel GA, Ogden RW (2010b) Modelling the layer-specific three-dimensional residual stresses in arteries, with an application to the human aorta. *J R Soc Interface* 7(46):787–799
- Holzapfel GA, Sommer G, Auer M, Regitnig P, Ogden RW (2007) Layer-specific 3D residual deformations of human aortas with non-atherosclerotic intimal thickening. *Ann Biomed Eng* 35(4):530–545
- Honda Y, Fitzgerald PJ (2008) Frontiers in intravascular imaging technologies. *Circulation* 117(15):2024–37
- Hsu M, Bazilevs Y (2011) Blood vessel tissue prestress modeling for vascular fluid–structure interaction simulation. *Finite Elem Anal Des* 47(6):593–599
- Hughes TJR, Liu WK, Zimmermann TK (1981) Lagrangian–Eulerian finite element formulation for incompressible viscous flows. *Comput Methods Appl Mech Eng* 29:329–349
- Hughes TJ, Franca LP, Hulbert GM (1989) A new finite element formulation for computational fluid dynamics: VIII. The Galerkin/least-squares method for advective–diffusive equations. *Comput Methods Appl Mech Eng* 73:173–189
- Humphrey JD (2002) *Cardiovascular solid mechanics: cells, tissues and organs*. Springer, Berlin
- Joosten MM, Dettmer WG, Perić D (2009) Analysis of the block Gauss–Seidel solution procedure for a strongly coupled model problem with reference to fluid–structure interaction. *Int J Numer Methods Eng* 78(7):757–778
- Kim H, Figueroa C, Hughes T, Jansen K, Taylor C (2009a) Augmented lagrangian method for constraining the shape of velocity profiles at outlet boundaries for three-dimensional finite element simulations of blood flow. *Comput Methods Appl Mech Eng* 198(45–46):3551–3566
- Kim HJ, Vignon-Clementel IE, Figueroa CA, LaDisa JF, Jansen KE, Feinstein JA, Taylor CA (2009b) On coupling a lumped parameter heart model and a three-dimensional finite element aorta model. *Ann Biomed Eng* 37(11):2153–2169
- Leiva J, Blanco P, Buscaglia G (2011) Partitioned analysis for dimensionally-heterogeneous hydraulic networks. *Multiscale Model Simul* 9(2):872–903
- Lewis M, Kerckhoffs R (2010) Current progress in patient-specific modeling. *Brief Bioinf* 11(1):111–126
- Li D, Robertson AM (2009) A structural multi-mechanism damage model for cerebral arterial tissue. *J Biomech Eng* 131(10):101–1013
- Lu J, Zhou X, Raghavan ML (2007) Inverse elastostatic stress analysis in pre-deformed biological structures: demonstration using abdominal aortic aneurysms. *J Biomech* 40(40):693–696
- Lu J, Zhou X, Raghavan M (2008) Inverse method of stress analysis for cerebral aneurysms. *Biomech Model Mechanobiol* 7(6):477–486
- Malossi ACI, Blanco PJ, Crosetto P, Deparis S, Quarteroni A (2013) Implicit coupling of one-dimensional and three-dimensional blood flow models with compliant vessels. *Multiscale Model Simul* 11(2):474–506
- Mendis S, Puska P, Norrving B (2011) *Global atlas on cardiovascular disease prevention and control*, chapter 1. World Health Organization, Geneva
- Migliavacca F, Balossino R, Pennati G, Dubini G, Hsia TY, de Leval MR, Bove EL (2006) Multiscale modelling in biofluidynamics: application to reconstructive paediatric cardiac surgery. *J Biomech* 39(6):1010–1020
- Mintz GS, Nissen SE, Anderson WD, Bailey SR et al (2001) American college of cardiology clinical expert consensus document on standards for acquisition, measurement and reporting of intravascular ultrasound studies (ivus) 33a report of the american college of cardiology task force on clinical expert consensus documents developed in collaboration with the european society of cardiology endorsed by the society of cardiac angiography and interventions. *J Am Coll Cardiol* 37(5):1478–1492
- Moireau P, Xiao N, Astorino M, Figueroa C, Chapelle D, Taylor CA, Gerbeau JF (2012) External tissue support and fluid–structure simulation in blood flows. *Biomech Model Mechanobiol* 11(1–2):1–18

- Orth RC, Wallace MJ, Kuo MD (2009) C-arm cone-beam CT: general principles and technical considerations for use in interventional radiology. *J Vasc Interv Radiol* 20(7):S538–S544
- Reymond P, Crosetto P, Deparis S, Quarteroni A, Stergiopoulos N (2013) Physiological simulation of blood flow in the aorta: comparison of hemodynamic indices as predicted by 3-D FSI, 3-D rigid wall and 1-D models. *Med Eng Phys* 35(6):784–791
- Rodriguez EK, Hoger A, McCulloch AD (1994) Stress-dependent finite growth in soft elastic tissues. *J Biomech* 27:455–467
- van Rooij WJ, Sprengers ME, de Gast AN, Peluso JPP, Sluzewski M (2008) 3D rotational angiography: the new gold standard in the detection of additional intracranial aneurysms. *AJNR Am J Neuroradiol* 29(5):976–979
- Southern J, Pitt-Francis J, Whiteley J, Stokeley D, Kobashi H, Nobes R, Kadooka Y, Gavaghan D (2008) Multi-scale computational modelling in biology and physiology. *Prog Biophys Mol Biol* 96(1–3):60–89
- Steinman DA, Thomas JB, Ladak HM, Milner JS, Rutt BK, Spence JD (2002) Reconstruction of carotid bifurcation hemodynamics and wall thickness using computational fluid dynamics and mri. *Magn Reson Med* 47(1):149–159
- Taylor C, Hughes T, Zarins C (1998) Finite element blood flow modeling in arteries. *Comput Methods Appl Mech Eng* 158:155–196
- Tezduyar TE, Takizawa K, Moorman C, Wright S, Christopher J (2010) Multiscale sequentially-coupled arterial FSI technique. *Comput Mech* 46(1):17–29
- Torii R, Oshima M, Kobayashi T, Takagi K, Tezduyar TE (2006) Computer modeling of cardiovascular fluid–structure interactions with the deforming-spatial-domain/stabilized space–time formulation. *Comput Methods Appl Mech Eng* 195(13–16):1885–1895
- Torii R, Oshima M, Kobayashi T, Takagi K, Tezduyar TE (2008) Fluid–structure interaction modeling of a patient-specific cerebral aneurysm: influence of structural modeling. *Comput Mech* 43(1):151–159
- Urquiza SA, Blanco PJ, Vénere MJ, Feijóo RA (2006) Multidimensional modeling for the carotid blood flow. *Comput Methods Appl Mech Eng* 195:4002–4017
- Valentín A, Humphrey JD (2009) Evaluation of fundamental hypotheses underlying constrained mixture models of arterial growth and remodelling. *Philos Trans Ser A Math Phys Eng Sci* 367(1902):3585–3606
- Vavourakis V, Papaharilaou Y, Ekaterinaris J (2011) Coupled fluid–structure interaction hemodynamics in a zero-pressure state corrected arterial geometry. *J Biomech* 44(13):2453–2460
- Vignon-Clementel IE, Alberto Figueroa C, Jansen KE, Taylor CA (2006) Outflow boundary conditions for three-dimensional finite element modeling of blood flow and pressure in arteries. *Comput Methods Appl Mech Eng* 195(29–32):3776–3796
- Watton PN, Selimovic A, Raberger NB, Huang P, Holzapfel GA, Ventikos Y (2011) Modelling evolution and the evolving mechanical environment of saccular cerebral aneurysms. *Biomech Model Mechanobiol* 10(1):109–132
- Wu Y, Cai XC (2014) A fully implicit domain decomposition based ALE framework for three-dimensional fluid–structure interaction with application in blood flow computation. *J Comput Phys* 258:524–537
- Zulliger MA, Fridez P, Hayashi K, Stergiopoulos N (2004) A strain energy function for arteries accounting for wall composition and structure. *J Biomech* 37:989–1000